Which Side Is Wrong? Neutral Contradiction Flags Do Not Help Retrieval-Augmented Generation Against Misinformation

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Abstract

Retrieval-augmented generation systems increasingly detect contradictions among retrieved passages and flag them to the generator. We ask what such a flag must say to help against misinformation. In experiments pre-registered in stages, on four benchmarks (including RAGuard's real misleading Reddit posts) and with seven generators from three model families, we vary the detector and the format of its flag. A perfectly accurate neutral flag ("these passages conflict") does not help: its pooled effect is -2.8 points $[-4.5, -1.0]$, and it significantly lowers accuracy in 12 of 34 settings. It changes *whether* the model answers, not *which side* it picks: when abstention is not allowed its effect vanishes, and its accuracy on answered questions is unchanged. Naming the false side helps on every benchmark and almost every generator, by about as much as stating the wrong answer outright. With real detectors the question becomes how often the detector is right where the generator is wrong. Across 30 crossed detector–generator pairs, a directional flag's gain rises with the detector's reliability relative to the generator (pre-registered, $\rho = +0.64$), and an exploratory accounting that charges each wrong-side flag against the questions the generator already answered correctly predicts the gain almost exactly ($\rho = 0.95$). Missing a conflict costs little; naming the wrong side is what hurts. We release the harness, the pre-registrations and every run.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al., 2020) hands a language model passages that may disagree. Some disagreements are genuine: two sources report different findings, or two people share a name. Others are planted or stale misinformation (Zou et al., 2025; Wang et al., 2025b). A natural fix is to detect the contradiction and tell the generator. Recent systems pair contradiction detectors with prompts that flag conflicting passages (Wang et al., 2026), and prompting a model about conflicts can help (Cattan et al., 2025). These systems report detection F1 and end-to-end accuracy, but not how the two connect. What should the flag say, and what must the detector get right for it to help?

We fix the retrieved passages and vary, independently, the *detector* (heuristic, NLI, LLM, random placebo, oracle, and synthetic detectors of chosen quality) and the *format* that passes its output to the generator: a neutral "these passages may conflict" flag, a flag that says one side is false without saying which, a *directional* flag that names the likely misinformation, or removal of the suspect passage. After pilots on RAMDocs (Wang et al., 2025b), we pre-registered our hypotheses and tested them on 300 held-out questions; six addenda, each committed before its runs, extended the study to ConflictBank (Su et al., 2024) (including a variant where the misinformation outnumbers the truth), to RAGuard (Zeng et al., 2025), whose misleading side is real Reddit posts, and to seven generators from OpenAI, Anthropic and Qwen. Controls separate what a flag conveys: a forced-choice prompt with no "unknown", flags stripped of their pair structure, an answer key that names the wrong answer without pointing at any passage, and a crossed design in which six detectors serve every generator. WikiContradict (Hou et al., 2024), whose conflicts are genuine, is the contrast case.

Our findings:

- **A correct neutral flag does not help against misinformation.** Pooled over 34 generator, benchmark and prompt settings, flagging exactly the true contradictions changes accuracy by -2.8 points $[-4.5, -1.0]$; it significantly lowers accuracy in 12 settings and raises it in 2. It moves answers into hedging or absten-

tion but does not change which side the model picks: under a forced-choice prompt its effect is null for three of four generators (H20), and accuracy on answered questions barely changes (H21). The same flag *raises* accuracy on WikiContradict, where both sides are valid.

- **Naming the false side is what helps.** The directional flag raises accuracy on every benchmark, for OpenAI, Anthropic and open models, including when the misinformation outnumbers the truth and on real Reddit posts. It does about as well as telling the model the wrong answer outright (H23), so what it conveys is the direction, not the location of the conflict. Saying that one side is false, without which, helps only when the other passages reveal which.
- **With a real detector, wrong-side flags are the cost.** Across 30 crossed detector–generator cells the directional flag's gain rises with the detector's reliability relative to the generator's (preregistered H24, $\rho = +0.64$, $p<0.001$). An exploratory accounting makes this precise: credit the detector for naming the false side where the generator would have been wrong, charge it for pointing at the correct passage where the generator would have been right ($\rho = 0.95$). Silence is cheap: the gpt-4o-mini detector marks a correct passage as suspect on 30.3% of questions and lowers the accuracy of the four generators most accurate on their own, while reasoning-model detectors flag less, rarely err, and lower no generator's.

For conflict-aware RAG, we recommend passing direction rather than a bare warning, letting the detector abstain, and measuring how often it names the wrong side on questions the generator already answers correctly. We release the harness, preregistrations and every run.[1]

2 Related Work

Knowledge conflicts. Language models can prefer memorised knowledge over contradicting context (Longpre et al., 2021), yet are readily persuaded by coherent contradicting evidence (Xie et al., 2024), to a degree that depends on the strength of their prior (Wu et al., 2024). Of the conflict types of Xu et al. (2024), we study inter-context conflicts, as in Chen et al. (2022), who find that models tend to rely on a single passage when evidence conflicts.

Benchmarks. RAMDocs (Wang et al., 2025b), built on AmbigDocs (Lee et al., 2024), mixes genuine ambiguity, misinformation and noise. CONFLICTS (Cattan et al., 2025) annotates search results with conflict types, each with its own desired behaviour, and WikiContradict (Hou et al., 2024) pairs genuinely contradicting Wikipedia passages. FaithEval (Ming et al., 2025), ConflictBank (Su et al., 2024) and ConfRAG (Yuan et al., 2026) cover related settings. RAMDocs and ConflictBank label which side is misinformation; WikiContradict does not.

Telling the model about conflicts. Prompting a model to reason about the conflict type helps, and supplying the gold type helps much more (Cattan et al., 2025); on WikiContradict, models acknowledge both sides more often when told the passages may conflict (Hou et al., 2024). Ragability (Gross et al., 2026) finds that such a hint helps on contradictory contexts but hurts on *consistent* ones, where it is false. In our neutral-flag result the hint is true and the input contradictory, and accuracy still falls, because the hint does not say which side to trust. Pan et al. (2023b) test detection and "vigilant" prompting against LLM-generated misinformation, and Pan et al. (2023a) call for misinformation-aware QA. Models weigh retrieved pages' metadata and appearance (Chiang and Lee, 2024), and source-credibility signals help a fact-checker resolve conflicting evidence (Ge et al., 2025). We separate the *content* of the hint from its *accuracy*.

Telling the model which passage is wrong. Closest to our directional format, Hong et al. (2024) pass a discriminator's marks on perturbed passages to the generator in the prompt, which makes it more robust to counterfactual noise. CrAM (Deng et al., 2025) rescales attention by passage credibility, RA-RAG (Hwang et al., 2025) estimates source reliability, and Weller et al. (2024) defend open-domain QA against disinformation. These works show that *some* reliability signal helps, but do not compare it with a neutral warning on the same pairs or measure its cost on conflict-free questions. RGB (Chen et al., 2024) warns models that passages may contain errors, and they still rarely detect them; our neutral flag is the pair-specific version of that warning.

[1] Code, the pre-registration and all run outputs are in the anonymised supplementary material and will be released publicly on acceptance.

Detection, resolution and poisoning. ConflictRAG (Wang et al., 2026) trains a pairwise detector with an LLM fallback and reports pair-level F1 alongside answer gains; its NLI and chain-of-thought baselines resemble two of our detectors. LLMs are imperfect context validators (Gokul et al., 2025), detect conflicts but struggle to pinpoint them (Wang et al., 2024), and tend to pick a side without explanation (Jiayang et al., 2024). They favour the more frequent side and the one matching their prior (Jin et al., 2024; Kortukov et al., 2024), prefer LLM-generated passages (Tan et al., 2024) (both ConflictBank sides are generated), are distracted by irrelevant context (Shi et al., 2023), and favour some sources (Schuster et al., 2026). TCR (Ye et al., 2026) uses a learned soft prompt, context-aware decoding (Shi et al., 2024) upweights the context, EvidentialRAG (Hossain et al., 2026) routes between answering and abstaining, and credibility-aware generation (Pan et al., 2024) fine-tunes models to weigh passage credibility, a trained analogue of our directional flags. MADAM-RAG (Wang et al., 2025b), Astute RAG (Wang et al., 2025a), Self-RAG (Asai et al., 2024), Corrective RAG (Yan et al., 2024) and InstructRAG (Wei et al., 2025) handle unreliable passages without an explicit contradiction detector; we compare against Astute RAG and TrustRAG. A few injected passages can steer a RAG system (Zou et al., 2025); RobustRAG (Xiang et al., 2024) isolates passages for certifiable robustness, and TrustRAG (Zhou et al., 2025) filters suspected poisoned ones, the analogue of our removal. We ask instead how a detector's output should be passed to the generator.

3 Framework

A question q comes with retrieved passages $P = (p_0, \ldots, p_{n-1})$. A *detector* returns flagged pairs $F \subseteq \{(i, j) : i < j\}$; a *directional* detector also returns, for each flagged pair, a *suspect* set $S_{ij} \subseteq \{i, j\}$: the side it believes is misinformation. A *format* turns (P, F, S) into the generator's prompt. We vary detectors and formats independently. Every prompt shares a skeleton: passages, question, and an instruction to end with a `Final answer(s):` line. The *default* skeleton asks for every valid answer when the passages support several (RAMDocs contains same-name ambiguity, so a correct answer may need several entities). The *minimal* skeleton drops that policy, and its answer line allows several answers without inviting them. The *single* skeleton says the question has exactly one correct answer (true of ConflictBank by construction). All three allow "unknown" when no passage answers the question; the *forced* skeleton is *single* without that option (Appendix A).

3.1 Formats

No flags (plain). Passages and question only.

Instruction. Plain plus a generic instruction to check the passages for contradictions, with no detector. v1 ends "if a disagreement reflects genuinely different valid answers, report each one"; v2 asks the model to judge which passages are reliable and to leave out answers supported only by unreliable ones.

Flag. Each flagged passage gets the header `[POTENTIAL CONFLICT with passage(s): j]`, followed by a list `Passages i and j: possible contradiction`. This is the original CAAG format, except that every detector's flags are rendered identically, so detectors differ only in *which* pairs they flag.

One-wrong flag. As Flag, but adds that exactly one passage of the pair contains misinformation, without saying which: it says that the conflict is misinformation, but not its direction.

Directional flag. As Flag, but suspect passages are headed `[LIKELY MISINFORMATION; conflicts with passage(s): j]`, and the list says which passage likely contains misinformation.

Controls. *Pick*: the neutral flag plus "decide which side is better supported and answer with that side only". *No pairs*: each flag without its pair structure (flagged passages are marked, but not which conflicts with which). *Answer key*: plain plus "The answer X is false" for each wrong answer, with no passage marked.

Remove (suspect). The suspect passages of every flagged pair are deleted. For a pair without a suspect, the second passage is deleted, as the original CAAG REMOVE strategy does without retrieval scores.

Single call. No detector: one call that first writes `Suspect passages: ...` and then answers from the other passages, at half the calls of a detector pipeline.

Closed book. No passages; it measures what the generator knows without retrieval.

3.2 Detectors

CAAG (Contradiction-Aware Attention Gating) is a heuristic detector from an earlier, unpublished RAG system, used with its code unchanged. It scores similar pairs by negation, entity-mismatch and temporal cues, runs locally and gives no direction (Appendix B).

NLI is `nli-deberta-v3-base` (He et al., 2023), a cross-encoder scored in both directions; a pair is flagged if the larger contradiction probability exceeds 0.5. It gives no direction.

LLM is a single call that lists contradicting pairs, telling them apart from ambiguity, complementary and temporal relations, and names the side more likely to be misinformation (or "both" or "unclear"), as JSON. It is gpt-4o-mini (temperature 0) unless stated; in the crossed design gpt-4.1, gpt-6-luna, gpt-6.1-terra, Claude Haiku 5.5 and Claude Sonnet 5.5 run the same prompt.

Random flags each pair independently at CAAG's per-pair flag rate, a placebo that matches CAAG's flag volume.

Oracle flags exactly the gold contradiction pairs and names the gold misinformation side. The *inverted oracle*, an adversarial control, flags the same pairs but names the correct side.

3.3 Synthetic detectors

To separate a detector's accuracy from its identity, we build detectors from the gold labels with a chosen pair-level precision $p \in \{0.25, 0.5, 0.75, 1\}$, recall $r \in \{0.25, 0.5, 1\}$ and direction accuracy $d \in \{0.5, 0.75, 1\}$ (36 cells), each rendered as a directional flag and as removal. All cells reuse the same random draws, so they are nested and comparisons across cells are paired (Appendix E).

4 Experimental Setup

Data. **RAMDocs** (Wang et al., 2025b) has 500 questions, each with 1–12 documents (5.5 on average) typed *correct* (supports a gold answer, possibly for a different same-name entity), *misinfo* (supports a wrong answer) or *noise*. A pair is a *contradiction* if one side is misinformation and the answers differ, *ambiguity* if both are correct with different answers, *consistent* if the answers match, and *unrelated* if either is noise. A question is a *conflict* question if it has a contradiction pair, and *clean* otherwise. Some documents typed *correct* contain a wrong answer; questions whose only sign of conflict is such a document count only in "All" (Appendix D).

ConflictBank (Su et al., 2024): for a Wikidata fact, an LLM writes one evidence passage for the true object and one for a substituted object. We sample 400 questions round-robin over relations: 300 conflict questions (correct passage, misinformation passage, two distractors from other subjects) and 100 clean ones (correct passage, three distractors). Every question has one gold answer. It had no pilot, so all its results are confirmatory. A *majority* variant gives each conflict question the correct passage and ConflictBank's three wrong-object passages for that fact (including its temporal and semantic conflicts), so the misinformation outnumbers the truth three to one.

RAGuard (Zeng et al., 2025) pairs PolitiFact claims (true or false) with Reddit posts that an LLM labelled supporting, misleading or unrelated, so the misleading side is real online text. We keep 294 claims with at least one supporting and one misleading post (up to two of each, plus two unrelated posts) and 100 claims with no misleading post; the question is whether the claim is true. Its labels are LLM-assigned stances, not verified facts.

WikiContradict (Hou et al., 2024) has 253 questions, each with two genuinely contradicting Wikipedia passages; the reference answer is *both* sides. We also run 237 automatically scorable **CONFLICTS** (Cattan et al., 2025) questions (Appendix D).

Pre-registration and splits. We developed the method on a random 200-question pilot split of RAMDocs. Before running anything on the remaining 300 questions (139 conflict, 126 clean, 35 excluded), we committed a pre-registration that fixes the conditions, metrics, hypotheses and corrections. Every RAMDocs result in the main text is on this confirmatory split; pilot results appear only in the appendix. Comparisons that were not registered, including every real-detector row of Table 3, are exploratory. Six addenda (H10–H26 and several re-analyses) were each committed before running anything they cover, so these hypotheses were formed after the original confirmatory results were known; the confirmatory core registered be-

fore any data is H1–H9. The repository's commit history is the only timestamp. Appendix D lists every deviation.

Generators and baselines. The main generator is `gpt-4o-mini` (OpenAI et al., 2024) (temperature 0.1, at most 256 output tokens, seeds 0–2). We replicate key conditions with `gpt-4.1`, Qwen2.5-7B-Instruct (Qwen Team et al., 2024) (4-bit, local), two current reasoning models, `gpt-6-luna` (3 seeds) and `gpt-6.1-terra` (seed 0), and Anthropic's Claude Haiku 5.5 (3 seeds) and Claude Sonnet 5.5 (seed 0). The reasoning and Claude models run at their default temperature with up to 4,096 output tokens, since they reason before answering. The LLM detector is gpt-4o-mini except in the crossed design (H24), where six detectors and six generators form 30 pairs on ConflictBank. On ConflictBank the primary prompt is the single skeleton; the default and minimal skeletons are robustness checks. We rerun the oracle and LLM-detector conditions with the minimal skeleton. MADAM-RAG (Wang et al., 2025b) is ported unchanged (three debate rounds). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) use their published prompts and defaults; since both return one answer, we also run *multi* variants that ask for every valid answer (Appendix D). We also run the LLM detector on Qwen2.5-7B with gpt-4o-mini generating. A detector pipeline counts as one detector call plus one generator call.

Metrics. On RAMDocs an answer is correct if it includes every gold answer and no wrong answer (Wang et al., 2025b). Our primary metric reads "includes" literally: after normalisation, each gold or wrong answer must appear as a whole-word span of the extracted final answer. A bidirectional variant, which also counts a predicted answer contained in a gold one, is the robustness metric. We chose the one-way metric before the confirmatory run because it is closer to MADAM-RAG's published score (Appendix D), and we compare conditions only within our own harness. On WikiContradict an answer is correct if it contains both sides (bidirectional matching); on ConflictBank and RAGuard we use the RAMDocs metrics. We also report the share of answers that contain the misinformation, the share that abstain ("unknown"), and accuracy among answered questions. An LLM judge (gpt-6.1-terra) agrees with the primary metric on 92.6% of 162 stratified conflict-question outputs ($\kappa = 0.85$); the metric is slightly conservative, missing some paraphrases.

Statistics. We average each question's score over seeds and report paired differences from the plain format with bootstrap 95% CIs that resample questions (10,000 resamples) and two-sided sign-flip permutation tests (Efron and Tibshirani, 1993; Dror et al., 2018) (10,000 permutations, extended to 200,000 when none reaches the observed statistic, so that a p-value can survive a large Holm family). CIs are unadjusted; significance is decided by the adjusted p below, so an interval can exclude zero for a difference that is not significant. A hypothesis is *robustly supported* if it also holds under the robustness metric. One significance scheme governs the paper: a registered test takes its family's Holm-adjusted p (Holm, 1979), and every other comparison we report (633 of them) forms one exploratory Holm family. Only results significant under this scheme are called significant or set in bold. A null difference is not read as equivalence without a TOST (Lakens, 2017). Appendix D gives the minimum detectable effect.

Cost. Token counts come from the API's usage reports. All experiments, pilots included, cost $169.23 in API fees (prices checked on 2026-10-27; Appendix D); Qwen ran locally.

5 Results

Table 3 (Appendix B) gives every RAMDocs condition, Table 2 every ConflictBank and RAGuard one, and Table 1 splits answers into correct, misinformation and abstention. Every pre-registered test is listed, by family, in the hypothesis tables of Appendix B.

5.1 A neutral flag changes whether the model answers, not what it answers

Flagging exactly the gold contradictions in the neutral format lowers gpt-4o-mini's RAMDocs conflict-question accuracy by 6.7 points [−10.8, −3.1] from 22.1% (H2), and gpt-4.1's by 8.6 (H9). On ConflictBank, even with a prompt stating that the question has one answer, it costs gpt-4o-mini 18.7 points and gpt-6-luna 9.4 (H15, H19). Pooled over all 34 settings (benchmarks, generators, prompts) with a random-effects model, the effect is −2.8 points [−4.5, −1.0] ($\tau = 4.9$; settings share questions, so the interval is optimistic); 12 settings are significantly harmed and

	No flags				Oracle, neutral flag				Oracle, directional flag			
Generator	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.
ConflictBank, one-answer prompt												
gpt-4o-mini	51.1	31.0	6.7	54.8	32.4	**17.7**	**41.0**	55.0	81.2	**4.2**	**2.1**	**83.0**
gpt-6-luna	53.6	24.3	11.9	60.8	44.1	**17.4**	**28.9**	62.0	90.0	**0.0**	**0.8**	**90.7**
Claude Haiku 5.5	66.8	22.0	1.2	67.6	67.4	22.1	0.9	68.0	90.1	**0.2**	1.1	**91.1**
gpt-6.1-terra	73.0	18.0	1.7	74.2	70.7	15.7	4.7	74.1	90.0	**0.7**	0.3	**90.3**
Claude Sonnet 5.5	75.7	15.7	1.0	76.4	78.7	12.0	1.7	80.0	92.0	1.0	1.7	93.6
gpt-4.1	64.7	27.3	1.7	65.8	56.7	21.7	14.0	65.9	91.3	0.3	1.7	92.9
Qwen2.5-7B	50.3	35.3	5.0	53.0	47.0	28.7	15.0	55.3	75.7	13.0	1.3	76.7
ConflictBank, forced choice												
gpt-4o-mini	55.4	31.6	0.1	55.5	55.2	31.9	0.4	55.5	79.7	**6.7**	0.0	**79.7**
gpt-6-luna	64.6	25.8	0.0	64.6	66.1	22.0	0.0	66.1	91.0	**0.1**	0.0	**91.0**
Claude Haiku 5.5	69.3	20.9	0.0	69.3	73.2	**16.8**	0.0	**73.2**	90.4	**0.0**	0.0	**90.4**
gpt-6.1-terra	77.7	14.7	0.0	77.7	77.3	13.3	0.0	77.3	92.3	**0.3**	0.0	**92.3**
RAGuard (real posts)												
gpt-4o-mini	36.7	31.2	32.1	54.1	39.9	27.8	32.3	59.0	46.6	**18.8**	34.6	**71.2**
gpt-6-luna	23.2	20.0	56.8	53.8	23.2	21.1	55.7	52.4	32.1	**9.8**	58.2	**76.7**
Claude Haiku 5.5	24.5	17.9	57.6	57.8	24.8	20.1	55.1	55.3	37.2	**11.0**	51.8	**77.2**
gpt-6.1-terra	24.8	14.3	60.9	63.5	27.2	16.7	56.1	62.0	31.6	**8.8**	59.5	78.2
Claude Sonnet 5.5	27.2	12.6	60.2	68.4	31.6	14.6	53.7	68.4	47.6	10.9	**41.5**	81.4
ConflictBank, misinformation majority												
gpt-4o-mini	21.6	64.1	3.2	22.3	22.7	**40.7**	**26.8**	**31.0**	54.2	**21.3**	**11.7**	**61.4**
gpt-6-luna	29.8	58.8	4.2	31.1	36.3	**43.6**	**11.7**	**41.1**	90.2	**0.2**	1.7	**91.8**
Claude Haiku 5.5	31.0	58.4	1.1	31.3	35.3	55.1	0.9	35.7	88.3	**1.0**	1.8	**89.9**
gpt-6.1-terra	26.0	65.3	1.0	26.3	22.3	71.0	0.7	22.5	92.3	**0.0**	1.3	**93.6**

Table 1: Risk and coverage on conflict questions (%): strictly correct answers, answers that contain the misinformation, abstentions ("unknown"), and selective accuracy (correct among non-abstained answers). **Bold**: the change from no flags is significant (H21 for ConflictBank; the paper's single scheme elsewhere). A neutral flag trades misinformation for abstention but never raises the share of correct answers; under forced choice it has no abstention to trade.

2 helped (gpt-6-luna under majority misinformation, Claude Haiku under the forced prompt). No Claude model is helped (H25: Haiku +0.7, Sonnet +3.0 on ConflictBank), and no generator on RAGuard (H26; 0.0 to +4.4, none significant).

The mechanism is abstention. Table 1 shows where the lost answers go. With the one-answer prompt, gpt-4o-mini's neutral flag lowers the share of answers containing the misinformation by 13.3 points, but only by abstaining (+34.3 points); its accuracy on the questions it still answers is unchanged (−1.0; H21). When "unknown" is not allowed (the forced prompt, H20), the neutral flag has no effect on gpt-4o-mini (−0.2), gpt-6-luna (+1.6) or gpt-6.1-terra (−0.3), and a small one on Claude Haiku (+3.9). Under the multi-answer prompt the flag instead makes the model list both sides (Figure 3). The answer policy decides how the model hedges; the directional flag cuts misinformation further while raising accuracy.

What the flag lacks is direction. A flag that says one side of each pair is false, without which, lowers gpt-4o-mini's ConflictBank accuracy by 35.7 points (H16). It helps only where the other passages reveal which side is false: on RAMDocs, where several correct passages usually agree, it gains gpt-6.1-terra +36.7 points, but on ConflictBank, one passage against one, −5.3. The pair structure does not leak direction: removing it from the one-wrong flag raises RAMDocs accuracy (H22: +6.5 with gpt-4o-mini, +10.1 with gpt-6-luna). Given an instruction to drop unreliable passages, adding the flag still costs 7.7 points on RAMDocs and 4.3 on ConflictBank (H13). Asking the model to commit to the better-supported side (*pick*) helps the reasoning models (+18.7 for gpt-6-luna on ConflictBank; exploratory). On WikiContradict, where both sides are valid, the neutral flag *raises* the share of answers giving both by 6.5 points [+3.0, +10.0] (H8).

5.2 Naming the false side is what helps

With the same oracle pairs, the directional flag beats the neutral and one-wrong flags in every setting we ran (H3, H9–H12, H15, H16, H19, H25, H26). On ConflictBank it gains between 17.0 and 36.4 points for the OpenAI generators, +23.3 for

			Accuracy		Oracle pairs				LLM detector		Single	LLM dir.
Generator	Prompt	Seeds	Closed	Plain	Flag	One wrong	Dir. flag	Remove	Flag	Dir. flag	call	clean q.
gpt-4o-mini	single	3	10.0	51.1	**−18.7**	**−35.7**	**+30.1**	**+35.0**	**−18.2**	+8.4	+7.6	−4.7
gpt-4o-mini	multi	3	–	21.6	**−8.8**	–	**+27.9**	**+66.4**	**−8.4**	**+20.0**	–	−1.0
gpt-4o-mini	minimal	3	–	35.0	**−18.1**	–	**+38.4**	**+51.2**	–	–	–	–
gpt-4.1	single	1	20.0	64.7	**−8.0**	**−35.3**	**+26.7**	**+26.0**	–	−2.3	–	−3.0
gpt-4.1	multi	1	–	13.0	**−6.3**	–	**+23.0**	**+78.3**	**−6.7**	**+17.0**	–	0.0
Qwen2.5-7B	single	1	4.3	50.3	−3.3	−5.0	**+25.3**	**+35.7**	−4.0	+7.3	–	−1.0
Qwen2.5-7B	multi	1	–	23.7	**−8.3**	–	**+17.7**	**+63.7**	−8.3	+6.7	–	−1.0
gpt-6-luna	single	3	16.9	53.6	**−9.4**	**−29.1**	**+36.4**	**+37.0**	**−9.3**	+8.6	−6.3	−1.3
gpt-6.1-terra	single	1	20.0	73.0	−2.3	−5.3	**+17.0**	**+18.3**	−2.7	−11.7	**−21.0**	0.0
Claude Haiku 5.5	single	3	11.8	66.8	+0.7	+6.6	**+23.3**	**+22.6**	+0.7	−4.1	+1.7	−5.0
Claude Sonnet 5.5	single	1	–	75.7	+3.0	+4.0	**+16.3**	–	–	−2.7	–	−1.0
gpt-4o-mini	forced	3	–	55.4	−0.2	**+7.6**	**+24.2**	–	–	+4.0	–	−2.7
gpt-6-luna	forced	3	–	64.6	+1.6	**+10.3**	**+26.4**	–	–	−2.4	–	−1.3
Claude Haiku 5.5	forced	3	–	69.3	**+3.9**	**+5.1**	**+21.1**	–	–	−4.7	–	−1.3
gpt-6.1-terra	forced	1	–	77.7	−0.3	**+6.7**	**+14.7**	–	–	**−14.3**	–	0.0
Majority misinformation: three wrong-side passages, one correct												
gpt-4o-mini	single	3	–	21.6	+1.1	−5.0	**+32.7**	**+65.1**	−2.2	**+5.7**	+6.8	−4.7
gpt-6-luna	single	3	–	29.8	**+6.6**	+0.3	**+60.4**	–	–	**+7.1**	–	−1.3
gpt-6.1-terra	single	1	–	26.0	−3.7	**+22.0**	**+66.3**	–	–	**+9.0**	–	0.0
Claude Haiku 5.5	single	3	–	31.0	+4.3	–	**+57.3**	–	–	**+6.8**	–	−5.0
RAGuard: real Reddit posts labelled supporting or misleading (294 conflict claims)												
gpt-4o-mini	single	3	64.9	36.7	+3.2	−5.7	**+9.9**	+5.9	–	+0.6	–	+1.7
gpt-6-luna	single	3	64.1	23.2	0.0	**−7.0**	**+8.8**	−1.0	–	+3.1	–	−1.3
Claude Haiku 5.5	single	3	65.1	24.5	+0.3	**+5.9**	**+12.7**	+2.0	–	**+5.0**	–	+1.3
gpt-6.1-terra	single	1	74.1	24.8	+2.4	+0.7	**+6.8**	−1.0	–	+2.0	–	+1.0
Claude Sonnet 5.5	single	1	–	27.2	+4.4	–	**+20.4**	–	–	**+7.1**	–	+10.0

Table 2: ConflictBank conflict questions (300; one correct answer each) and RAGuard conflict claims (294; answer true or false): accuracy (%, one-way strict) closed-book and with the passages but no flags, then the paired difference in points made by each format. *forced*: the one-answer prompt without the option to answer "unknown". *Prompt*: the answer policy: *single* says there is one answer, *multi* asks for every valid answer (our default skeleton), *minimal* states no policy. *One wrong*: the neutral flag plus "exactly one of the two contains misinformation". *Single call*: one call that names the suspect passages, then answers. Last column: the LLM directional flag's change on the 100 clean questions. The LLM detector is gpt-4o-mini for every generator. **Bold**: significant under the paper's single scheme.

Claude Haiku and +16.3 for Claude Sonnet. It gains under the forced prompt (H20: +14.7 to +26.4), when the misinformation outnumbers the truth three to one (+32.7 to +66.3; Claude Haiku +57.3), and on RAGuard's real posts (H26: +6.8 to +20.4, all significant). On RAMDocs it gains 13.4 points with gpt-4o-mini and up to 39.6 with gpt-6.1-terra, but not significantly for Qwen2.5-7B (Table 13). It raises accuracy on answered questions by +15.9 to +30.8 points (H21): unlike the neutral flag, it changes which side the model picks.

On RAGuard every generator is less accurate with the posts than closed-book (gpt-4o-mini 36.7% against 64.9%), as Zeng et al. (2025) report, mostly because the stronger models abstain (gpt-6-luna on 56.8%); the flag removes misinformation answers but not this abstention.

Direction works like an answer key. Telling the model "the answer X is false", without marking any passage, does about as well as the directional flag: none of the nine registered comparisons differs significantly (H23; e.g. +5.3 on ConflictBank and −1.7 on RAMDocs for gpt-4o-mini). The oracle rows are therefore upper bounds on what any detector can deliver, and the value of a flag lies in its direction. A wrong direction is as strong a lever the other way: an inverted oracle that names the correct side costs 7.0 points on RAMDocs and 8.6 on ConflictBank (exploratory).

5.3 With real detectors, wrong-side flags are the cost

A real detector's directional flag replaces the generator's judgement with the detector's, so it should pay only when the detector's is better. H24 tests this on a common scale across 30 crossed cells (Figure 1): the gain rises with the detector's direction accuracy minus the generator's no-flag accuracy ($\rho = +0.64$, $p<0.001$; within cells, 0.48 points of gain per point of gap, [0.43, 0.52]). It

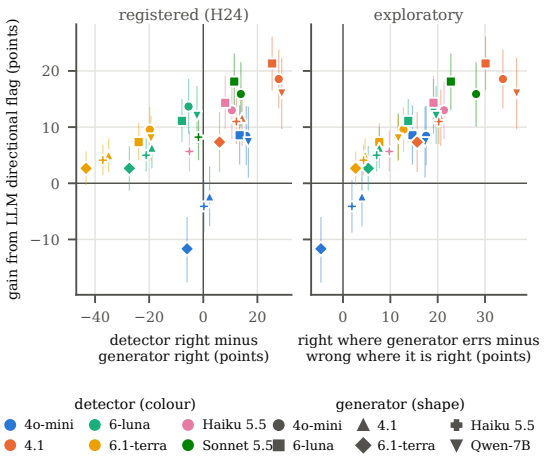


Figure 1: Each point is a detector–generator cell (ConflictBank, one-answer prompt, 300 conflict questions; bars: 95% CIs). Left: the registered predictor, the detector's direction accuracy minus the generator's no-flag accuracy. Right: the exploratory score, which credits right flags only where the generator errs and charges wrong-side flags only where it is right. Table 22 lists the cells.

holds on the cells registered before Claude was added ($\rho = +0.60$, $p = 0.003$) and is weaker under majority misinformation ($\rho = +0.30$, $p = 0.132$; exploratory), where the gpt-4o-mini detector's direction falls to 49.4%.

Yet the gain is positive in 27 of 30 cells, including many where the detector is less accurate than the generator, so "the detector must beat the generator" is too strong. An exploratory, post-hoc accounting explains why. A flag can help only where the generator would be wrong, and hurt only where it would be right, so we credit the detector for naming the false side on the first kind of question and charge it for marking a correct passage on the second. This score predicts the cell gains far better than the gap ($\rho = 0.95$; Pearson $r = 0.90$ against 0.52). The gpt-4o-mini detector flags almost every question (97.7%) and marks a correct passage as suspect on 30.3% of them: it helps the weaker generators (gpt-4o-mini +8.4, Qwen2.5-7B +7.3) and lowers the accuracy of the four most accurate (gpt-6.1-terra −11.7; also Claude Sonnet, Claude Haiku and gpt-4.1). gpt-6.1-terra as a detector flags only 37.7% of questions and marks a correct passage on 4.0%; it names the false side less often than gpt-4o-mini does, yet lowers no generator's accuracy. On RAGuard the same detector helps the Claude generators (H26: Haiku +5.0, Sonnet +7.1) but not gpt-4o-mini (+0.6).

Against published methods (H14) and a single call (H18). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) lose legitimate same-name answers on RAMDocs (gold recall 69.0% to 47.9% and 29.4%), and on ConflictBank they do not beat the one-answer prompt (−2.1 and −9.7 points; exploratory). A single call that names the suspect passages before answering does not differ significantly from the gpt-4o-mini detector pipeline, at half the calls (H18; equivalence within ±3 points is not established), but the reasoning models and Claude Haiku, asked to judge ConflictBank's generated passages, distrust most of them and abstain, losing up to 51.0 points on clean questions. Table 21 (Appendix C) gives all baselines.

5.4 What a detector must get right

In synthetic detectors the gain rises with direction accuracy, and precision matters mainly for removal, whose false flags delete correct passages (H4a); the grid predicts real detectors' effects only roughly (H6). Appendix E gives the cells and tests.

6 Discussion

Why neutral flags do not help. A neutral flag says that two passages disagree but not which to trust, so the model falls back on its answer policy: it lists both sides, abstains, or, when it must answer, picks as it would have without the flag. For misinformation, the useful part of knowing the conflict type (Cattan et al., 2025) is its direction.

What a detector should optimise. Because a directional flag works like an answer key, a detector's direction is only as good as its errors are rare. Its value on a question depends on the generator: naming the false side helps where the generator would have been misled, and naming the true side harms where it would not. Detection F1 and direction accuracy hide this; the detectors that served every generator well stayed silent when unsure.

Recommendations. Pass direction rather than a bare warning. Let the detector abstain, and measure, on held-out conflicts, its wrong-side rate where the target generator is right on its own: a detector good enough for a small generator can hurt a larger one. Gate removal on its measured clean-question loss (20.6 points with our LLM detector).

Limitations

Models. We test seven generators from three families: five OpenAI models (including two current reasoning models), Claude Haiku 5.5 and Sonnet 5.5, and Qwen2.5-7B at 4-bit precision. We could not run Gemini, and no large open model. Qwen2.5-7B, gpt-6.1-terra and Claude Sonnet ran with one seed, and the reasoning and Claude models only at their default temperature, so their sampling is not deterministic, and Claude ran through Anthropic's OpenAI-compatible endpoint. The crossed reliability test (H24) covers 30 detector–generator cells on one benchmark; its mechanism score was chosen after seeing the cells and needs confirmation on new data. The LLM detector is a single prompt. The published baselines are reimplemented (Astute RAG has no official code), and we do not compare against a trained detector such as ConflictRAG's (Wang et al., 2026).

Data and power. The confirmatory evidence comes from two benchmarks: RAMDocs (300 questions, 139 of them conflict questions) and ConflictBank (400 sampled questions, 300 with conflicts). In both, the misinformation is synthetic, made by editing or generating text around a substituted answer; ConflictBank's passages are LLM-written on both sides. RAGuard's misleading side is real, but its stance labels are assigned by an LLM, and with its posts in context most generators abstain on most claims, which limits what flags can show there. RAMDocs' document labels are imperfect. We exclude 35 holdout questions whose status depends on documents we could not label without human annotation, and we check that the conclusions survive treating those documents as misinformation. Our comparisons detect effects of about 5.8 points with 80% power, so smaller effects, including small harms on clean questions, may be missed. WikiContradict has genuine conflicts but no misinformation side. CONFLICTS shows no significant effect for any condition; scored only on items with a reference answer, it can neither confirm nor refute our findings. The majority variant tests one case where the false side is more numerous, but its extra wrong-side passages are generated for the same substitution, which a real attacker would vary. The current reasoning models distrust ConflictBank's generated passages when asked to judge them (Section 5.3), which suggests that generated benchmarks may not measure how such models treat real documents.

Metric. RAMDocs ships no scorer. Our primary metric reconstructs the published definition and scores our MADAM-RAG port close to the published number on the pilot split, but the published subset is unknown, so this is not a reproduction. The metric is string matching: it misses paraphrases, and partial answers interact with "no wrong answer". An LLM judge agrees with it on 92.6% of sampled outputs, but no human labels check either. We report a more lenient variant throughout and mark any conclusion that depends on the choice.

Direction and the grid. Oracle direction does as well as an explicit answer key (H23), so the oracle rows are upper bounds. In the synthetic grid the gain rises with direction accuracy, but the grid cannot separate adjacent levels. The real LLM detector names the right side 88.3% of the time on RAMDocs and 68.8% on ConflictBank. Its directional flag's gain over all ConflictBank questions with gpt-4o-mini is not significant after correction (H15), and with stronger generators it hurts. Synthetic false flags are spread uniformly and their suspect side is random, whereas real detectors concentrate their errors. The grid predicts real detectors only roughly, and three detectors cannot show whether document- or pair-level summaries predict better.

Pre-registration. The original hypotheses (H1–H9) were registered before the confirmatory run. H10–H26 were registered in six addenda, after the earlier results were known though before their own runs, and the only timestamp is the repository's commit history, not a third-party registry. Several analyses were chosen after seeing data and are labelled as such: the TOST margin, the single- and multi-answer split, the document-level placement axes, the post-hoc comparison of detector and generator reliability that H24 later tested, H24's mechanism score, and the refinement of permutation p-values at their floor.

Scope. We study flags passed to the generator as text. Learned interfaces such as soft prompts or credibility fine-tuning (Ye et al., 2026; Pan et al., 2024), and retrieval-side filtering, may follow different trade-offs. We test one adversarial case, a detector that always names the wrong side, but not an attacker who controls which pairs are flagged.

Ethics Statement

This work uses public benchmarks and commercial model APIs, and involves no human subjects or personal data. Its main risk is dual use. Our results show that a false "likely misinformation" label, or the removal of a passage, can change a RAG system's answers: an inverted oracle that names the correct side as misinformation costs up to 21.8 points. An adversary who can trigger a deployed detector could exploit this (Zou et al., 2025). We therefore report harm on conflict-free questions alongside every gain, and we argue that systems should measure what a detector does at the document level before acting on its flags.

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A Prompts

All prompts below are copied verbatim from the code that produced the results.

Generation skeleton (default).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Minimal skeleton (no answer policy).

```
Answer the question using the passages below.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your answer; if there is more than
one, separate them with |>
```

Single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Closed book (single-answer version).

```
Answer the question from your own knowledge. The
question has exactly one correct answer.
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Instruction v1.

```
Some passages may contradict each other. Before
answering, check whether they do.
Ignore passages that appear to contain
misinformation, and if a disagreement reflects
genuinely different valid answers, report each one.
```

Instruction v2.

```
Some passages may contain false information. Before
answering, decide which passages are reliable by
checking them against each other.
Do not include an answer that is supported only by a
passage you judge to be unreliable.
```

Single call (inserted as the instruction).

```
Some passages may contain misinformation. First
decide which passages, if any, contain
misinformation, by checking them against each other
and against what you know, and write one line in
this format:
Suspect passages: <passage numbers, or none>
Then answer the question using only the other
passages.
```

Example: directional flag (passage 2 suspect).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION; conflicts with
passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
```

```
- Passages 0 and 2: possible contradiction; passage
2 likely contains misinformation

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Example: one-wrong flag, single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [CONFLICT with passage(s): 2; one side
contains misinformation]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [CONFLICT with passage(s): 0; one side
contains misinformation]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; exactly
one of the two contains misinformation (it is not
known which)

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Directional LLM detector.

```
You will see a question and {n} numbered passages
retrieved for it.
Some passages may contain misinformation: false
claims about the answer to the question.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

For each contradiction, say which passage of the
pair is more likely the misinformation, judging
by agreement with the other passages and by
plausibility: its number, "both" if both are likely
false, or "unclear" if you cannot tell.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation":
"contradiction|ambiguity|complementary|temporal",
"misinformation": <int>|"both"|"unclear"}}]}}
Use the passage numbers shown. "misinformation" is
needed only for contradictions. Return {{"pairs":
[]}} if there are none.
```

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	29.9	31.3	22.1	41.5	911
Prompt only					
Instruction (v1)	+1.6	+0.8	−1.2	+2.9	948
Instruction (v2)	+3.1	+2.3	+5.3	−1.1	951
Oracle detector					
Oracle, flag	−3.1	−3.5	**−6.7**	0.0	980
Oracle, directional flag	+6.2	+7.0	**+13.4**	0.0	998
Oracle, remove suspect	**+12.8**	**+14.5**	**+27.6**	0.0	825
Real detectors (exploratory)					
Random, flag	+1.0	+1.0	+0.7	+1.3	949
CAAG, flag	−0.4	−0.8	−0.7	−0.8	942
CAAG, remove	−3.3	−3.8	0.0	−7.9	809
NLI, flag	+0.2	+0.3	−0.5	+1.1	1021
NLI, remove	−5.7	−5.0	−2.2	−8.2	592
LLM, flag	−1.8	−1.3	−4.1	+1.9	970
LLM, directional flag	+5.0	+4.9	+12.5	−3.4	983
LLM, remove suspect	+0.3	0.0	+18.7	**−20.6**	718
Reference					
MADAM-RAG	+5.7	+5.7	+3.6	+7.9	8196

Table 3: RAMDocs hold-out, gpt-4o-mini, 3 seeds, one-way strict accuracy (primary). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

B Additional Results

Robustness to the metric and the labels. Table 15 repeats Table 3 with the bidirectional metric. Table 16 repeats the pre-registered tests with every suspect document treated as misinformation. Table 4 lists all pre-registered tests.

The heuristic detector. CAAG scores a pair only if its MiniLM (Reimers and Gurevych, 2019) cosine similarity reaches $\theta_s = 0.65$. Its score is the sum of a negation and contrast asymmetry term, a named-entity mismatch term (differing dates, numbers, money or percentages) and a temporal-marker term, multiplied by the similarity. A pair is flagged if the score reaches $\theta_c = 0.35$. Its per-pair flag rate on the confirmatory split (6.22%) sets the rate of the random detector. CAAG is the detector we set out to study. As a neutral flag it did not beat random flags of the same volume, and as a removal rule it only deleted correct passages (−7.9 points on clean questions, 0.0 on conflict ones). Its false flags fall mostly on pairs that should both be reported: 134 of its 221 false flags on the hold-out are same-name ambiguity pairs, which its entity-

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H1: detection quality					
H1	F1 llm_dir > nli	+2.6 [−2.4, +7.5]	0.311	not supported	–
H1	F1 nli > caag	+12.2 [+7.5, +16.9]	<0.001	supported	–
H1	direction accuracy llm_dir > 50% (cluster CI[r])	88.3% [83.3, 92.9]	<0.001	supported	–
H2: a correct neutral flag					
H2	oracle+flag < plain (conflict)	−6.7 [−10.8, −3.1]	<0.001	supported	yes
H3: what direction adds					
H3	oracle+flag_directional > plain (conflict)	+13.4 [+6.7, +20.4]	<0.001	supported	yes
H3	oracle+flag_directional > oracle+flag (conflict)	+20.1 [+13.9, +26.9]	<0.001	supported	yes
H3	oracle+remove_suspect > plain (conflict)	+27.6 [+20.6, +35.0]	<0.001	supported	yes
H4: false flags					
H4a	false-flag cells below plain on clean q., directional flag[r]	0 of 9 cells	Holm	not supported	yes
H4a	false-flag cells below plain on clean q., removal[r]	6 of 9 cells	Holm	supported	yes
H4b	harm grows with clean-q. exposure, Spearman over 18 cells[r]	$\rho = -0.39$ [−0.74, −0.12]	–	supported	yes
H4c	removal harms clean q. more than directional flag[r]	5 of 9 matched cells	Holm	count only	–
H5: the heuristic detector vs random					
H5	caag+flag vs random+flag (all)	−1.4 [−3.4, +0.3]	0.178	n.s.	yes
H5	equivalence within ±3 points (TOST)[r]	90% CI [−3.1, +0.1]	–	not equivalent	yes
H6: does the grid predict real detectors					
H6	actual effect inside the 95% interval of the document-level prediction[r]	7 of 8 (pair level: 6 of 8)	–	descriptive	–
H6	document-level error < pair-level error (paired, exact sign-flip)[r]	−1.2 [−3.6, +0.4]	0.414	not supported	yes
H7: prompt-only instructions					
H7	instruct vs plain (conflict)	−1.2 [−4.3, +1.7]	1.000	n.s.	yes
H7	instruct vs plain (clean)	+2.9 [0.0, +6.1]	0.280	n.s.	yes
H7	instruct_v2 vs plain (conflict)	+5.3 [+1.4, +9.6]	0.059	n.s.	yes
H7	instruct_v2 vs plain (clean)	−1.1 [−5.0, +2.6]	1.000	n.s.	yes
H9: gpt-4.1					
H9	gpt-4.1: oracle+flag < plain (conflict)	−8.6 [−13.2, −4.3]	<0.001	supported	yes
H9	gpt-4.1: oracle+flag_directional > plain (conflict)	+18.2 [+12.0, +24.7]	<0.001	supported	yes
H9	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+26.9 [+20.1, +33.6]	<0.001	supported	yes
H9	gpt-4.1: oracle+remove_suspect > plain (conflict)	+30.2 [+22.5, +37.9]	<0.001	supported	yes
H8: genuine conflicts (WikiContradict)					
H8	WikiContradict: instruct vs plain	+8.7 [+5.4, +12.3]	<0.001	significant	–
H8	WikiContradict: instruct_v2 vs plain	−6.7 [−10.0, −3.7]	<0.001	significant	–
H8	WikiContradict: oracle+flag vs plain	+6.5 [+3.0, +10.0]	<0.001	supported	–
H8	WikiContradict: caag+flag vs plain	+0.7 [0.0, +1.6]	0.250	n.s.	–
H8	WikiContradict: nli+flag vs plain	+3.2 [+0.3, +6.1]	0.077	n.s.	–
H8	WikiContradict: llm_dir+flag vs plain	+6.2 [+2.8, +9.6]	<0.001	significant	–

Table 4: Pre-registered tests registered before the confirmatory run (RAMDocs hold-out unless stated; one-way strict accuracy; differences in points). Rows marked [r] are re-analyses registered later, before they were computed. Directional predictions are *supported* or *not supported*; tests with no predicted sign are *significant* or *n.s.* *Robust*: the same outcome under the bidirectional metric. p-values come from 10,000 sign-flip permutations (smallest attainable raw p is $1/10{,}001$), Holm-adjusted within the family; this is the significance used everywhere in the paper.

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H10: without the answer policy (minimal skeleton)					
H10	gpt-4o-mini: oracle+flag vs plain (conflict)	-5.3 [-9.4, -1.4]	0.042	significant	yes
H10	gpt-4o-mini: oracle+flag_directional > plain (conflict)	$+5.0$ [-1.7, $+11.8$]	0.314	not supported	yes
H10	gpt-4o-mini: oracle+flag_directional > oracle+flag (conflict)	$+10.3$ [$+4.8$, $+16.1$]	0.005	supported	yes
H10	gpt-4o-mini: oracle+remove_suspect > plain (conflict)	$+18.9$ [$+12.5$, $+25.9$]	<0.001	supported	yes
H10	gpt-4.1: oracle+flag vs plain (conflict)	-11.0 [-17.5, -4.6]	0.010	significant	yes
H10	gpt-4.1: oracle+flag_directional > plain (conflict)	$+12.7$ [$+5.8$, $+19.7$]	0.005	supported	yes
H10	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	$+23.7$ [$+17.0$, $+30.7$]	<0.001	supported	yes
H10	gpt-4.1: oracle+remove_suspect > plain (conflict)	$+14.9$ [$+8.6$, $+21.6$]	<0.001	supported	yes
H10	Qwen2.5-7B: oracle+flag vs plain (conflict)	-5.8 [-10.8, -1.4]	0.117	n.s.	yes
H10	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+2.9$ [-3.6, $+9.4$]	0.509	not supported	yes
H10	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+8.6$ [$+2.9$, $+14.4$]	0.041	supported	yes
H10	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+10.8$ [$+4.3$, $+17.3$]	0.017	supported	yes
H11: an open-weights generator					
H11	Qwen2.5-7B: oracle+flag < plain (conflict)	-5.8 [-10.8, -0.7]	0.114	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+3.6$ [-0.7, $+8.6$]	0.224	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+9.4$ [$+5.0$, $+14.4$]	0.001	supported	yes
H11	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+18.0$ [$+10.8$, $+25.2$]	<0.001	supported	yes

Table 5: Pre-registered tests added in the first addendum (H10–H11), each committed before the runs it covers. Same conventions as Table 4. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H12: ConflictBank					
H12	gpt-4o-mini: neutral flag vs plain (conflict)	−8.8 [−12.8, −4.9]	<0.001	significant	yes
H12	gpt-4o-mini: dir. flag > plain (conflict)	+27.9 [+22.8, +32.9]	<0.001	supported	yes
H12	gpt-4o-mini: dir. flag > neutral flag (conflict)	+36.7 [+31.6, +41.9]	<0.001	supported	yes
H12	gpt-4o-mini: removal > plain (conflict)	+66.4 [+61.1, +71.7]	<0.001	supported	yes
H12	gpt-4.1: neutral flag vs plain (conflict)	−6.3 [−9.7, −3.3]	<0.001	significant	yes
H12	gpt-4.1: dir. flag > plain (conflict)	+23.0 [+18.0, +28.3]	<0.001	supported	yes
H12	gpt-4.1: dir. flag > neutral flag (conflict)	+29.3 [+24.0, +34.7]	<0.001	supported	yes
H12	gpt-4.1: removal > plain (conflict)	+78.3 [+73.3, +83.0]	<0.001	supported	yes
H12	Qwen2.5-7B: neutral flag vs plain (conflict)	−8.3 [−12.3, −4.3]	<0.001	significant	yes
H12	Qwen2.5-7B: dir. flag > plain (conflict)	+17.7 [+12.0, +23.3]	<0.001	supported	yes
H12	Qwen2.5-7B: dir. flag > neutral flag (conflict)	+26.0 [+20.7, +31.3]	<0.001	supported	yes
H12	Qwen2.5-7B: removal > plain (conflict)	+63.7 [+58.3, +69.0]	<0.001	supported	yes
H13: neutral flag + resolution instruction					
H13	RAMDocs, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	−2.4 [−6.2, +1.2]	0.743	n.s.	yes
H13	RAMDocs, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+15.6 [+9.6, +21.8]	<0.001	supported	yes
H13	RAMDocs, gpt-4.1: neutral flag + instruction vs plain (conflict)	+5.8 [−2.9, +14.4]	0.743	n.s.	yes
H13	RAMDocs, gpt-4.1: dir. flag + instruction > neutral flag + instruction (conflict)	+25.2 [+18.0, +32.4]	<0.001	supported	yes
H13	ConflictBank, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	+1.6 [−2.8, +6.0]	0.743	n.s.	yes
H13	ConflictBank, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+62.2 [+57.1, +67.2]	<0.001	supported	yes
H14: published baselines					
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs astute multi (labelled)	+10.8 [+5.9, +16.0]	<0.001	significant	yes
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs trustrag multi (labelled)	+23.0 [+18.1, +28.3]	<0.001	significant	yes
H14	RAMDocs, gpt-4.1: LLM dir. flag vs astute multi (labelled)	+6.4 [+0.4, +12.5]	0.151	n.s.	no
H14	RAMDocs, gpt-4.1: LLM dir. flag vs trustrag multi (labelled)	+19.6 [+13.6, +25.7]	<0.001	significant	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs astute (all questions)	−3.4 [−8.1, +1.3]	0.324	n.s.	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs trustrag (all questions)	+0.8 [−3.9, +5.6]	0.761	n.s.	yes

Table 6: Pre-registered tests added in the second (H12–H14), each committed before the runs it covers. Same conventions as Table 4. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H15: ConflictBank, single-answer prompt					
H15	ConflictBank, gpt-4o-mini: neutral flag vs plain (conflict)	−18.7 [−23.4, −14.0]	<0.001	significant	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > neutral flag (conflict)	+48.8 [+43.6, +54.1]	<0.001	supported	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > plain (conflict)	+30.1 [+25.1, +35.2]	<0.001	supported	yes
H15	ConflictBank, gpt-4o-mini: LLM dir. flag > plain (all q.)	+5.2 [+1.0, +9.3]	0.065	not supported	no
H15	ConflictBank, gpt-4.1: neutral flag vs plain (conflict)	−8.0 [−12.7, −3.7]	0.006	significant	yes
H15	ConflictBank, gpt-4.1: dir. flag > neutral flag (conflict)	+34.7 [+29.0, +40.3]	<0.001	supported	yes
H15	ConflictBank, gpt-4.1: dir. flag > plain (conflict)	+26.7 [+21.7, +31.7]	<0.001	supported	yes
H15	ConflictBank, gpt-4.1: LLM dir. flag > plain (all q.)	−2.5 [−6.5, +1.5]	0.522	not supported	yes
H15	ConflictBank, Qwen2.5-7B: neutral flag vs plain (conflict)	−3.3 [−9.0, +2.0]	0.522	n.s.	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > neutral flag (conflict)	+28.7 [+23.3, +34.0]	<0.001	supported	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > plain (conflict)	+25.3 [+19.7, +31.0]	<0.001	supported	yes
H15	ConflictBank, Qwen2.5-7B: LLM dir. flag > plain (all q.)	+5.2 [+0.5, +10.0]	0.130	not supported	yes
H16: information vs direction (one-wrong flag)					
H16	ConflictBank, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+65.8 [+60.6, +70.9]	<0.001	supported	yes
H16	ConflictBank, gpt-4o-mini: one-wrong flag vs plain (conflict)	−35.7 [−41.1, −30.3]	<0.001	significant	yes
H16	ConflictBank, gpt-4.1: dir. flag > one-wrong flag (conflict)	+62.0 [+56.0, +67.7]	<0.001	supported	yes
H16	ConflictBank, gpt-4.1: one-wrong flag vs plain (conflict)	−35.3 [−41.3, −29.3]	<0.001	significant	yes
H16	RAMDocs, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+20.6 [+14.4, +27.3]	<0.001	supported	yes
H16	RAMDocs, gpt-4o-mini: one-wrong flag vs plain (conflict)	−7.2 [−12.2, −2.6]	0.006	significant	yes
H17: majority misinformation					
H17	CB majority, gpt-4o-mini: neutral flag vs plain (conflict)	+1.1 [−2.7, +4.9]	0.603	n.s.	yes
H17	CB majority, gpt-4o-mini: dir. flag > plain (conflict)	+32.7 [+27.3, +37.9]	<0.001	supported	yes
H17	CB majority, gpt-4o-mini: LLM dir. flag vs plain (conflict)	+5.7 [+1.4, +9.9]	0.017	significant	yes
H18: a single-call baseline					
H18	RAMDocs, gpt-4o-mini: LLM dir. flag vs self-identify (labelled)	−1.3 [−4.8, +2.1]	1.000	n.s.	yes
H18	ConflictBank, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−0.2 [−4.2, +3.7]	1.000	n.s.	yes
H18	CB majority, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−1.8 [−4.6, +1.1]	0.730	n.s.	yes
H19: current reasoning models					
H19	ConflictBank, gpt-6-luna: neutral flag vs plain (conflict)	−9.4 [−12.9, −6.0]	<0.001	significant	yes
H19	ConflictBank, gpt-6-luna: dir. flag > neutral flag (conflict)	+45.9 [+40.8, +50.9]	<0.001	supported	yes
H19	ConflictBank, gpt-6-luna: dir. flag > one-wrong flag (conflict)	+65.6 [+60.7, +70.1]	<0.001	supported	yes
H19	ConflictBank, gpt-6-luna: LLM dir. flag > plain (all q.)	+6.1 [+2.1, +10.0]	0.019	supported	yes
H19	ConflictBank, gpt-6.1-terra: neutral flag vs plain (conflict)	−2.3 [−6.7, +2.0]	0.717	n.s.	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+19.3 [+14.7, +24.0]	<0.001	supported	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > one-wrong flag (conflict)	+22.3 [+17.7, +27.3]	<0.001	supported	yes
H19	ConflictBank, gpt-6.1-terra: LLM dir. flag > plain (all q.)	−8.8 [−13.2, −4.2]	0.002	not supported	yes
H19	RAMDocs, gpt-6-luna: neutral flag vs plain (conflict)	−3.1 [−5.3, −1.4]	0.007	significant	yes
H19	RAMDocs, gpt-6-luna: dir. flag > neutral flag (conflict)	+42.2 [+34.5, +49.9]	<0.001	supported	yes
H19	RAMDocs, gpt-6.1-terra: neutral flag vs plain (conflict)	−0.7 [−2.9, +1.4]	1.000	n.s.	yes
H19	RAMDocs, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+40.3 [+32.4, +48.2]	<0.001	supported	yes
H19	CB majority, gpt-6-luna: dir. flag > plain (conflict)	+60.4 [+55.3, +65.4]	<0.001	supported	yes
H19	CB majority, gpt-6-luna: LLM dir. flag vs plain (conflict)	+7.1 [+2.4, +11.8]	0.019	significant	yes

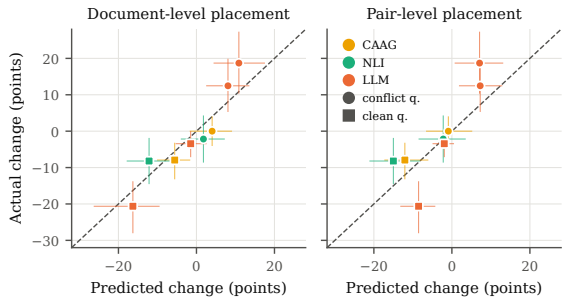


Figure 2: Effect of real detectors predicted from the synthetic grid (x, with the 95% bootstrap prediction interval) against the measured effect (y, with its 95% CI), for conflict (•) and clean (■) questions. Left: detectors located by their document-level behaviour; right: by pair precision, recall and direction accuracy.

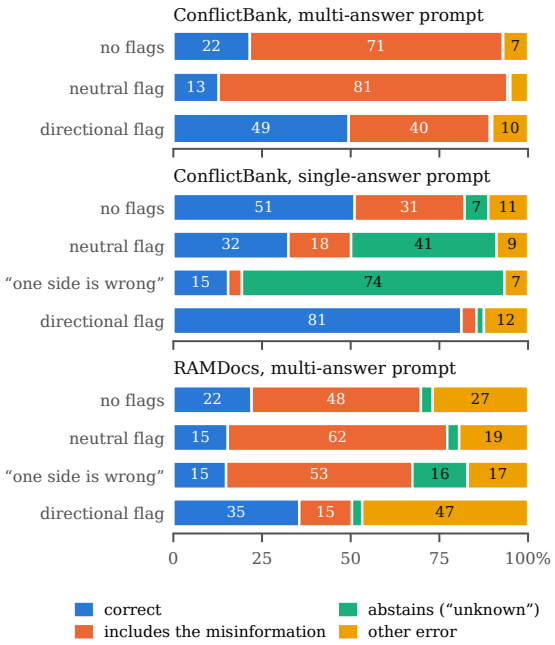


Figure 3: What a flag does to answers on conflict questions (gpt-4o-mini, oracle pairs; share of answers, %). The answer policy decides how a neutral flag fails: a model told to give every valid answer adds the misinformation, and one told there is one answer abstains. Only the directional flag raises the share of correct answers.

mismatch signal flags because they mention different dates or numbers. We read this as a warning about lexical contradiction cues in general (Gururangan et al., 2018), rather than as a problem with a particular threshold. Retuning illustrates why F1 misleads. Over a 121-setting sweep, CAAG's best pair F1 is 29.2, above NLI's, at $\theta_s = 0.40$ and $\theta_c = 0.10$. But it gets there by flagging 43% of all pairs and 87% of clean questions, which our grid predicts would be harmful under removal (Appendix B).

Original CAAG format. Rendering CAAG's flags in its original format, with typed lines and confidence scores ("factual conflict (confidence: 0.41)"), changes accuracy by −0.9 points overall (−1.2 on conflict questions, −1.6 on clean ones). This is no different from the uniform format.

Pilot study (exploratory). On the 200-question pilot split, scored with the bidirectional metric, the same pattern appeared before any confirmatory run. Oracle neutral flags changed conflict-question accuracy by −10.5 points, directional flags by +11.8, and removing the suspect by +28.4. The pilots also showed that the original REMOVE strategy, which deletes an arbitrary side of each pair, removes correct documents. They are why we added directional formats, synthetic direction accuracy and document-level placement. All pilot runs are released.

Phase-0 audit of CAAG on NarrativeQA. CAAG was designed for long-document QA. On 100 NarrativeQA (Kočiský et al., 2018) questions from 50 stories (BM25 top-10 chunks from a single coherent story), it flags at least one pair in 54% of questions, 166 pairs in all. A single story rarely contradicts itself, so most of these flags are likely false; the flagged pairs are released for annotation. This is the setting where a false flag can only hurt.

C Further Results

C.1 Where the directional flag fails (exploratory)

After seeing the minimal-skeleton run, we split conflict questions by the number of gold answers. 105 of 139 have several (same-name ambiguity *and* misinformation), and a strict answer must list all of them. On the 34 single-answer conflict questions, the directional flag's estimate is positive in every setting, from +11.8 points (Qwen) to +46.1 (gpt-4o-mini), both with the default skeleton. On multi-answer questions it changes little except with gpt-4.1 (gpt-4o-mini +2.9, and −5.7 with the minimal skeleton; gpt-4.1 +19.0). One reading is that a warning about one passage makes the smaller models more conservative about the legitimate alternative answers too; we did not test this. Removal does not depend on the model acting on a warning, and it gains in both groups.

C.2 Real detectors

CAAG flags 34.1% of clean questions. Its neutral flags do not beat random flags at the same rate (H5; -0.4 vs. $+1.0$ points overall). The 90% CI of the difference, $[-3.1, +0.1]$, rules out a gain larger than $+0.1$ points but does not establish equivalence within ± 3 points (TOST). The directional-versus-neutral comparison with the LLM detector's pairs was registered as exploratory in the addendum. Under the bidirectional metric, the LLM detector's directional flag costs clean questions 6.1 points. With gpt-4.1 as the generator, its gain over all 300 questions ($+5.4$) is significant after Holm, but the 35 unlabelled questions contribute disproportionately to it (they gain 13.3 points); over the labelled questions it is $+4.4$ $[+0.6, +8.3]$. On ConflictBank the NLI cross-encoder flags no pair at all: ConflictBank passages are long, LLM-written and agree on everything but the one fact, and the truncated pair rarely contains it.

C.3 A detector from another family

Replacing the gpt-4o-mini detector with Qwen2.5-7B (gpt-4o-mini still generates) gives a much less selective detector. It flags 96.8% of clean RAMDocs questions, but its direction accuracy is 89.0%. Its directional flag gains $+10.6$ points on conflict questions and 0.0 on clean ones, and its neutral flag -2.6. Removal again trades a conflict-question gain ($+14.1$) for a clean-question loss (-14.6). The conclusions do not depend on the detector's family.

C.4 CONFLICTS

On the 237 automatically scorable CONFLICTS questions (exploratory, one seed; most have no conflict), no condition moves accuracy significantly (Table 18).

D Reproducibility

Runs. Every number in this paper is produced by one script from committed run outputs: per-call responses, token counts, detector outputs and configurations. The pre-registration was committed before any confirmatory run. Its deviations are listed below.

Deviations from the pre-registration.

1. *Primary metric.* The first draft of the pre-registration named the bidirectional metric as primary. Before the confirmatory run we switched to one-way matching, because it reproduces MADAM-RAG's published score on the pilot split (Section 4); bidirectional became the robustness metric. Every table reports both.
2. *Second generator.* The plan named an open-weights model. None was available before the confirmatory run, so gpt-4.1 was used (H9). We later added Qwen2.5-7B-Instruct, registered before its run. Because local inference takes about 3 s per call, the Qwen runs use one seed; we made that change before inspecting any Qwen result.
3. *CONFLICTS judge.* The plan called for a judge from a different model family, validated on about 100 human labels. Neither the judge nor the labels were available, so CONFLICTS is scored only on items with a reference answer, with one seed, and reported as exploratory.
4. *H4.* The pre-registration counted 54 false-flag cells. On clean questions, cells that differ only in direction accuracy render identical prompts (their scores differ by exactly 0), so they are one comparison: 18 in all, 9 per format. We report H4a per format. The cell-permutation p-value for the dose-response correlation assumed exchangeable cells; the cells are nested, so we replace it with a question-level bootstrap CI. No threshold was set for H4c, so we report a count.
5. *H5.* A non-significant difference is not equivalence. We add a TOST with a ± 3-point margin, chosen in the addendum after the confirmatory run, so it is post hoc.
6. *H6.* The first draft tested whether the grid prediction lies inside the actual effect's CI, which is lenient. We now report the pre-registered test (actual effect inside a bootstrap prediction interval) and a paired test of the two spaces' errors.
7. *Placement modes.* For H6, the detectors without direction were run under the label `remove_suspect` as well as `remove`. Both render identical prompts, so these were served entirely from the response cache.
8. *Significance.* Earlier drafts used different corrections in different tables. We now apply one scheme throughout (Section 4), which makes some exploratory differences that were bold in earlier drafts non-significant.
9. *Single- and multi-answer split.* The split of conflict questions by number of gold answers

(Appendix C.1) was chosen after seeing the minimal-skeleton run and is exploratory.

10. *Second registration.* We later registered H12–H14 (ConflictBank, the neutral flag with a resolution instruction, and the published baselines), the Qwen2.5-7B detector and the ConflictBank sampling rule. The gpt-4.1 runs in it use seed 0 only, to stay within budget. One ConflictBank gpt-4o-mini run stopped on a transient API error and was resumed from the response cache; no row was re-sampled.

11. *Third and fourth registrations.* H15–H18 (§11) and H19 (§12) were committed before their runs. To stay within budget, §12 dropped two §11 runs before they started: gpt-4.1 under majority misinformation (replaced by gpt-6.1-terra in H17) and gpt-4.1 seeds 1–2, so H14 with gpt-4.1 stays on seed 0 as first registered and every gpt-4.1 baseline comparison is seed-0 matched.

12. *Reasoning models.* gpt-6-luna and gpt-6.1-terra accept only their default temperature and require `max_completion_tokens`; we used their defaults with a 4,096-token budget. 25 responses from the two models (almost all closed-book) were empty because reasoning used the whole budget; they are scored as wrong.

13. *Fifth and sixth registrations.* H20–H24 (§13) and H25–H26 with Claude and RAGuard (§14) were committed before their runs. The planned `cross_mj` runs completed within budget. Gemini was planned but not run, because no API key was available. Claude rejects a temperature and JSON mode on the endpoint we used; it ran at its default temperature, and as a detector its JSON is parsed by the detector's regex fallback. The answer parser unwraps paired tags (`<answer>X</answer>`), which no earlier result contained.

14. *Permutation resolution.* With 10,000 permutations the smallest p-value is 10^{-4}, which cannot survive Holm over 633 exploratory comparisons. After the §14 runs, and before writing them up, we extended the permutations to 200,000 whenever none of the first 10,000 reached the observed statistic. This changes only p-values at their floor, and only lowers them.

15. *H24 additions.* The cross-cell break-even point, the count of positive cells and the mechanism score (credit for naming the false side where the generator errs, charge for marking a correct passage where it is right) were defined after seeing the H24 cells and are exploratory.

16. *Run interruptions.* Several runs were stopped and resumed (a transient API error; reordering the run queue). Resumption keeps completed rows and re-requests only missing ones; no row was re-sampled.

The addenda (H10–H26 and the re-analyses above) and the Qwen seed change were each committed to the repository before the runs they cover. The repository's commit history is the only timestamp; we did not use a third-party registry (van Miltenburg et al., 2021).

RAMDocs labels. The released document types are imperfect: 163 documents typed *correct* contain a wrong answer, some as injected misinformation, others incidentally. We leave pairs that touch them out of pair-level metrics, and questions whose only sign of conflict is such a document (55 of 500; 35 of the confirmatory split) count only in "All". Treating them as misinformation is a robustness check.

Power. With 139 RAMDocs conflict questions, the minimum detectable effect at 80% power is 4–12 points (median 5.8; Card et al., 2020).

Metric choice. With the one-way metric, our port of MADAM-RAG scores 28.5% on the pilot split and 32.8% on all 500 questions (published: 28.00, on an undisclosed subset); with the bidirectional metric it scores 38.4% on all 500. Because the published subset and scorer are unknown, we do not claim to reproduce the published number. Normalisation lower-cases and removes punctuation and articles.

WikiContradict and CONFLICTS. A WikiContradict answer is correct if both of the dataset's answer strings (one per passage) match a final answer under the bidirectional rule. From CONFLICTS we use the 237 questions that have a reference answer (outdated-information, misinformation and no-conflict items); an answer is correct if it contains the reference answer.

Baselines. MADAM-RAG is ported unchanged to our client: three debate rounds, temperature 0, 128 tokens per call. Astute RAG has no official code. We use the prompts in its ap-

pendix verbatim with its default settings (t=1 consolidation round, m=1 generated passage, temperature 0, 1,024 output tokens). A generated passage that says "I don't know" is dropped, and the answer is the text in its `<ANSWER>` tag. TrustRAG is ported from its repository (commit `11dcea0`, MIT licence): SimCSE CLS embeddings (`sup-simcse-bert-base-uncased`), a ROUGE-L gate of 0.25, 2-means clustering and a cosine threshold of 0.88, followed by its three prompts verbatim at temperature 0.1. Its system message is not sent, because our client sends user messages only. The *multi* variants change one sentence of the final step to ask for every valid answer (Astute RAG) or append our final-answer line (TrustRAG); the unmodified variants are also reported.

ConflictBank. We use the QA file of the dataset's Hugging Face release (revision `92866a0`). Questions are sampled round-robin over relations with seed 0; distractors are other subjects' correct passages. The sampling script, the item ids and the seed are released; the data are not redistributed, because the release states a licence in its README but ships no licence file.

RAGuard. We use the Hugging Face release `UCSC-IRKM/RAGuard` (revision `dae344a2`, MIT licence on the card). A post is usable if it has at least 15 words after removing the "[Link Post]" placeholder and is not a duplicate within its claim; posts are cut to 300 words and shuffled by claim id. Supporting $\times$ misleading pairs are the contradictions and misleading posts the misinformation side. The build script and item ids are released.

Cost and compute. All API calls in this paper, pilots included, cost $169.23. Prices were checked on 2026-10-27 (per million input, cached-input and output tokens: gpt-4o-mini $0.15, $0.075 and $0.60; gpt-4.1 $2.00, $0.50 and $8.00; Claude Haiku 5.5 $0.10, $0.01 and $0.50; Claude Sonnet 5.5 $2.00, $0.10 and $10.00). CAAG and NLI ran on a laptop CPU (0.2 s and 1.2 s per question).

E Synthetic Detectors and the Grid

Construction. Each synthetic detector (Section 3.3) is built from the gold labels. Each gold contradiction pair is flagged with probability r. False flags are drawn from non-contradicting pairs at a single per-pair rate, set over the whole split so

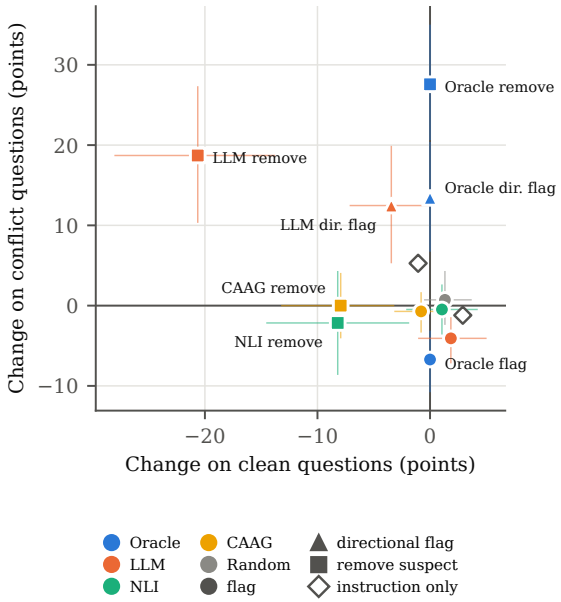


Figure 4: Change in one-way strict accuracy on RAMDocs conflict questions (y) against clean questions (x), relative to no flags (gpt-4o-mini, default skeleton, confirmatory split, 95% CIs). Colour: detector; shape: format. The oracle sits on the y-axis because it never flags a clean question. Without direction, removal deletes the second passage of each pair.

that expected precision is p. That rate is dataset-wide, so conflict-free questions receive false flags too. A true flag names the gold misinformation side with probability d, and otherwise the other side. A false flag has no misinformation side, so its suspect is picked uniformly. Each pair gets one uniform draw for flagging and one for direction, and every cell reuses them. Cells are therefore nested: raising r or d, or lowering p, only adds flags or corrections. This keeps comparisons across cells paired.

Results. Figure 5 replaces the oracle with synthetic detectors of known quality. Gains rise with direction accuracy in both formats. Only the cells with perfect direction are significant on all questions (directional flag +6, removal +13 points at perfect precision), but neighbouring cells at direction accuracy 0.75 have similar point estimates on conflict questions, so this is a dose-response, not a threshold: with 36 cells under Holm, the grid has little power to separate adjacent levels. Precision matters mainly for removal. On clean questions a cell's prompt does not depend on direction accuracy, so the false-flag cells reduce to 9 distinct comparisons per format. Removal is significantly harmful in 6 of them, the directional flag in

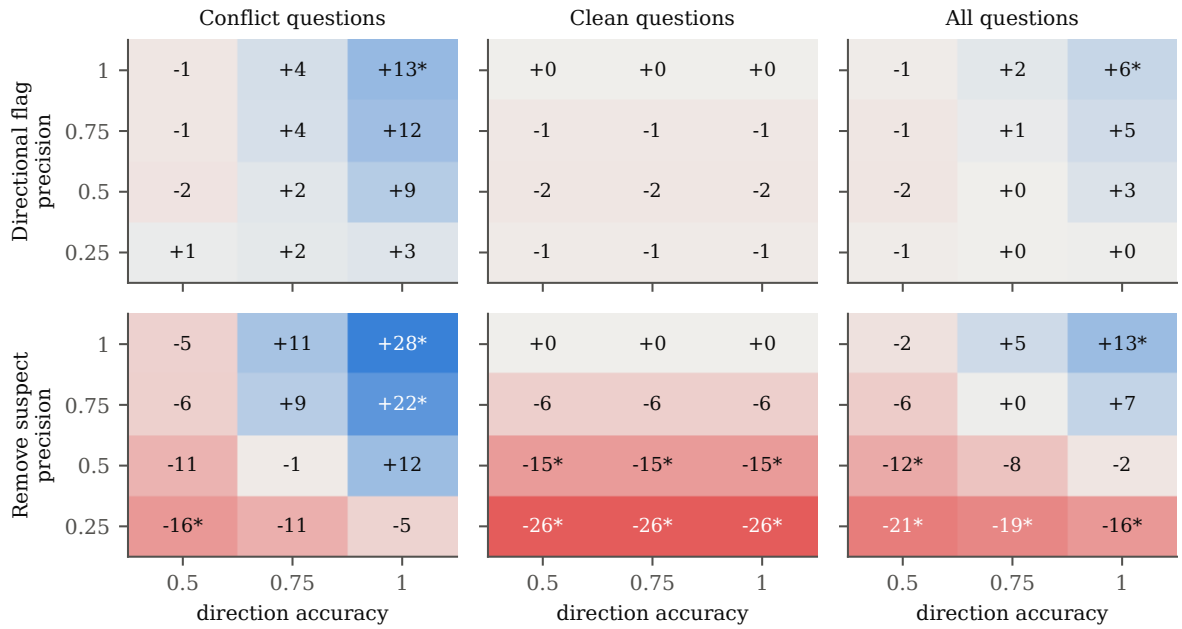


Figure 5: Synthetic detectors at recall 1: change in one-way strict accuracy (points) against no flags, by pair precision (rows) and direction accuracy (columns), for the directional flag (top) and removal (bottom). *: significant after Holm correction over the 36 cells of that mode and subset. Clean questions only see false flags, so their rows do not vary with direction accuracy.

0 (H4a): false flags make the model delete correct passages, which costs clean questions 15 points at precision 0.5 and 26 at 0.25, while synthetic false directional flags cost at most two points. The harm tracks the share of clean questions that lose a correct document (Spearman ρ =-0.39 over 18 distinct cells, 95% CI [−0.74, −0.12] from a question-level bootstrap; H4b). Removal hurt clean questions significantly more than the directional flag in 5 of 9 matched cells (H4c; we set no threshold, so we report the count). Removal is the more powerful and the more fragile interface: its best cell gains twice the directional flag's on all questions, but below precision 0.75 it is the only format that does net harm.

Placing real detectors on the grid. Can the grid predict a real detector's effect from a few summary numbers? We compare two ways of summarising a detector. *Pair-level*: its pair precision, recall and direction accuracy. *Document-level*: what it does to each subset's documents in a given format. For conflict questions, that is the share whose misinformation documents are all marked and the share that have a correct document marked; for clean questions, the share that have a correct document marked. Here "marked" means removed (Remove) or headed as misinformation (Directional flag). The prediction for a subset is the inverse-distance-weighted mean effect of the four nearest grid cells in that space.

Does the grid predict real detectors? If the grid only described synthetic detectors, it would be of limited use. Figure 2 (Appendix B) tests whether it predicts real ones (H6). As preregistered, we bootstrap the whole prediction (cell effects, every detector's location and the interpolation) over questions, and ask whether the actual effect falls inside the 95% interval. Locating detectors by what they do to documents (the share of conflict questions whose misinformation is marked, and the share of questions that have a correct document marked) puts 7 of 8 effects inside; locating them by pair precision, recall and direction accuracy puts 6 inside. Document-level errors are smaller on average (4.1 vs. 5.3 points), but the paired difference, −1.2 [−3.6, +0.4] (exact sign-flip $p = 0.41$), is not significant over 8 cases. Both spaces miss the same case, the LLM detector's removal gain on conflict questions: one correctly directed pair is enough to remove a misinformation document, so the detector removes the misinformation on 61.2% of conflict questions despite its low pair recall. We chose the document-level axes after seeing the LLM detector on the

pilot split. The grid is a useful first estimate of a detector's effect, but with three detectors we cannot claim that one summary predicts better than the other.

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H19: current reasoning models					
H19	CB majority, gpt-6.1-terra: dir. flag > plain (conflict)	+66.3 [+61.0, +71.7]	<0.001	supported	yes
H19	CB majority, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	+9.0 [+3.3, +14.7]	0.019	significant	yes

Table 8: Pre-registered tests added in the third (H15–H19) (continued), each committed before the runs it covers. Same conventions as Table 4. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H20: forced choice (no "unknown")					
H20	ConflictBank forced, gpt-4o-mini: neutral flag vs plain (conflict)	-0.2 [-4.1, $+3.7$]	1.000	n.s.	yes
H20	ConflictBank forced, gpt-4o-mini: one-wrong flag vs plain (conflict)	$+7.6$ [$+2.7$, $+12.3$]	0.024	significant	yes
H20	ConflictBank forced, gpt-4o-mini: dir. flag > plain (conflict)	$+24.2$ [$+19.7$, $+29.0$]	<0.001	supported	yes
H20	ConflictBank forced, gpt-6-luna: neutral flag vs plain (conflict)	$+1.6$ [-1.6, $+4.8$]	1.000	n.s.	yes
H20	ConflictBank forced, gpt-6-luna: one-wrong flag vs plain (conflict)	$+10.3$ [$+5.6$, $+15.1$]	<0.001	significant	yes
H20	ConflictBank forced, gpt-6-luna: dir. flag > plain (conflict)	$+26.4$ [$+21.9$, $+31.0$]	<0.001	supported	yes
H20	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain (conflict)	-0.3 [-4.0, $+3.3$]	1.000	n.s.	yes
H20	ConflictBank forced, gpt-6.1-terra: one-wrong flag vs plain (conflict)	$+6.7$ [$+2.3$, $+11.3$]	0.034	significant	yes
H20	ConflictBank forced, gpt-6.1-terra: dir. flag > plain (conflict)	$+14.7$ [$+10.3$, $+19.0$]	<0.001	supported	yes
H20	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain (conflict)	$+3.9$ [$+1.2$, $+6.7$]	0.034	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: one-wrong flag vs plain (conflict)	$+5.1$ [$+1.4$, $+8.8$]	0.034	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: dir. flag > plain (conflict)	$+21.1$ [$+16.9$, $+25.4$]	<0.001	supported	yes
H22: pair topology					
H22	RAMDocs, gpt-4o-mini: one-wrong without pairs vs with pairs (conflict)	$+6.5$ [$+2.4$, $+10.8$]	0.015	significant	yes
H22	RAMDocs, gpt-6-luna: one-wrong without pairs vs with pairs (conflict)	$+10.1$ [$+5.8$, $+14.9$]	<0.001	significant	yes
H22	RAMDocs, gpt-6.1-terra: one-wrong without pairs vs with pairs (conflict)	-0.7 [-6.5, $+5.8$]	1.000	n.s.	yes
H22	CB majority, gpt-4o-mini: neutral without pairs vs with pairs (conflict)	-3.1 [-6.4, $+0.2$]	0.390	n.s.	yes
H22	CB majority, gpt-4o-mini: LLM dir. flag without pairs vs with pairs (conflict)	$+0.3$ [-2.1, $+2.8$]	1.000	n.s.	yes
H22	CB majority, gpt-6-luna: neutral without pairs vs with pairs (conflict)	-6.1 [-9.1, -3.1]	0.002	significant	yes
H22	CB majority, gpt-6-luna: LLM dir. flag without pairs vs with pairs (conflict)	$+0.8$ [-1.7, $+3.1$]	1.000	n.s.	yes
H22	CB majority, gpt-6.1-terra: neutral without pairs vs with pairs (conflict)	$+10.0$ [$+5.7$, $+14.3$]	0.002	significant	yes
H22	CB majority, gpt-6.1-terra: LLM dir. flag without pairs vs with pairs (conflict)	$+1.7$ [-1.3, $+4.7$]	1.000	n.s.	yes
H23: answer key					
H23	RAMDocs, gpt-4o-mini: dir. flag vs answer key (conflict)	-1.7 [-6.2, $+3.1$]	1.000	n.s.	yes
H23	RAMDocs, gpt-6-luna: dir. flag vs answer key (conflict)	-4.1 [-9.1, $+0.7$]	0.808	n.s.	yes
H23	RAMDocs, gpt-6.1-terra: dir. flag vs answer key (conflict)	-5.0 [-10.8, $+0.7$]	0.827	n.s.	yes
H23	ConflictBank, gpt-4o-mini: dir. flag vs answer key (conflict)	$+5.3$ [$+1.6$, $+9.1$]	0.059	n.s.	no
H23	ConflictBank, gpt-6-luna: dir. flag vs answer key (conflict)	$+0.1$ [-1.6, $+1.8$]	1.000	n.s.	yes
H23	ConflictBank, gpt-6.1-terra: dir. flag vs answer key (conflict)	-1.7 [-3.3, -0.3]	0.515	n.s.	yes
H23	CB majority, gpt-4o-mini: dir. flag vs answer key (conflict)	$+3.2$ [-1.8, $+8.2$]	0.905	n.s.	yes
H23	CB majority, gpt-6-luna: dir. flag vs answer key (conflict)	$+0.4$ [-1.2, $+2.2$]	1.000	n.s.	yes
H23	CB majority, gpt-6.1-terra: dir. flag vs answer key (conflict)	$+2.3$ [$+0.3$, $+4.7$]	0.515	n.s.	yes
H25: Claude					
H25	ConflictBank, claude-haiku-5-5: neutral flag vs plain (conflict)	$+0.7$ [-2.2, $+3.4$]	1.000	n.s.	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+22.7$ [$+18.6$, $+26.9$]	<0.001	supported	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > one-wrong flag (conflict)	$+16.8$ [$+13.0$, $+20.8$]	<0.001	supported	yes
H25	ConflictBank, claude-haiku-5-5: LLM dir. flag > plain (all q.)	-4.3 [-8.1, -0.7]	0.145	not supported	yes
H25	ConflictBank, claude-sonnet-5-5: neutral flag vs plain (conflict)	$+3.0$ [-0.3, $+6.3$]	0.634	n.s.	yes
H25	ConflictBank, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+13.3$ [$+9.7$, $+17.3$]	<0.001	supported	yes

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H25: Claude					
H25	RAMDocs, claude-haiku-5-5: neutral flag vs plain (conflict)	+1.2 [−2.6, +5.3]	1.000	n.s.	yes
H25	RAMDocs, claude-haiku-5-5: dir. flag > neutral flag (conflict)	+27.1 [+20.1, +34.1]	<0.001	supported	yes
H25	RAMDocs, claude-sonnet-5-5: neutral flag vs plain (conflict)	0.0 [−4.3, +4.3]	1.000	n.s.	yes
H25	RAMDocs, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	+14.4 [+7.9, +21.6]	0.002	supported	yes
H25	CB majority, claude-haiku-5-5: dir. flag > plain (conflict)	+57.3 [+52.0, +62.7]	<0.001	supported	yes
H25	CB majority, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	+6.8 [+2.1, +11.4]	0.043	significant	yes
H26: RAGuard (real misleading posts)					
H26	RAGuard, gpt-4o-mini: neutral flag vs plain (conflict)	+3.2 [+0.1, +6.3]	0.426	n.s.	yes
H26	RAGuard, gpt-4o-mini: dir. flag > plain (conflict)	+9.9 [+5.6, +14.2]	<0.001	supported	yes
H26	RAGuard, gpt-4o-mini: dir. flag > neutral flag (conflict)	+6.7 [+2.9, +10.5]	0.008	supported	yes
H26	RAGuard, gpt-4o-mini: one-wrong flag vs plain (conflict)	−5.7 [−10.2, −1.2]	0.161	n.s.	yes
H26	RAGuard, gpt-4o-mini: LLM dir. flag vs plain (conflict)	+0.6 [−2.8, +4.0]	1.000	n.s.	yes
H26	RAGuard, gpt-6-luna: neutral flag vs plain (conflict)	0.0 [−1.7, +1.7]	1.000	n.s.	yes
H26	RAGuard, gpt-6-luna: dir. flag > plain (conflict)	+8.8 [+5.7, +12.1]	<0.001	supported	yes
H26	RAGuard, gpt-6-luna: dir. flag > neutral flag (conflict)	+8.8 [+5.7, +12.2]	<0.001	supported	yes
H26	RAGuard, gpt-6-luna: one-wrong flag vs plain (conflict)	−7.0 [−10.3, −3.9]	<0.001	significant	yes
H26	RAGuard, gpt-6-luna: LLM dir. flag vs plain (conflict)	+3.1 [−0.1, +6.2]	0.511	n.s.	yes
H26	RAGuard, gpt-6.1-terra: neutral flag vs plain (conflict)	+2.4 [0.0, +5.1]	0.731	n.s.	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > plain (conflict)	+6.8 [+3.4, +10.5]	0.005	supported	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+4.4 [+1.4, +7.8]	0.126	not supported	yes
H26	RAGuard, gpt-6.1-terra: one-wrong flag vs plain (conflict)	+0.7 [−3.4, +4.8]	1.000	n.s.	yes
H26	RAGuard, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	+2.0 [−1.0, +5.1]	1.000	n.s.	yes
H26	RAGuard, claude-haiku-5-5: neutral flag vs plain (conflict)	+0.3 [−2.0, +2.7]	1.000	n.s.	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > plain (conflict)	+12.7 [+9.2, +16.3]	<0.001	supported	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > neutral flag (conflict)	+12.4 [+9.0, +15.9]	<0.001	supported	yes
H26	RAGuard, claude-haiku-5-5: one-wrong flag vs plain (conflict)	+5.9 [+2.9, +9.0]	0.005	significant	yes
H26	RAGuard, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	+5.0 [+1.8, +8.2]	0.025	significant	yes
H26	RAGuard, claude-sonnet-5-5: neutral flag vs plain (conflict)	+4.4 [+0.7, +8.2]	0.321	n.s.	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > plain (conflict)	+20.4 [+15.6, +25.2]	<0.001	supported	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	+16.0 [+11.6, +20.4]	<0.001	supported	yes
H26	RAGuard, claude-sonnet-5-5: LLM dir. flag vs plain (conflict)	+7.1 [+3.1, +11.2]	0.022	significant	yes
H21: risk and coverage					
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	−13.3 [−17.6, −9.1]	<0.001	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	−26.8 [−31.6, −22.0]	<0.001	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	−6.9 [−9.9, −4.0]	<0.001	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	−24.3 [−28.6, −20.2]	<0.001	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	−2.3 [−6.0, +1.3]	1.000	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	−17.3 [−21.7, −13.0]	<0.001	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	+0.1 [−2.6, +2.8]	1.000	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	−21.8 [−26.1, −17.6]	<0.001	significant	–

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	+0.3 [−3.4, +4.2]	1.000	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	−24.9 [−29.7, −20.1]	<0.001	significant	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	−3.8 [−6.8, −0.9]	0.079	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	−25.7 [−30.0, −21.3]	<0.001	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	−1.3 [−5.0, +2.3]	1.000	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	−14.3 [−18.3, −10.3]	<0.001	significant	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	−4.1 [−6.9, −1.3]	0.025	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	−20.9 [−25.1, −16.8]	<0.001	significant	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+34.3 [+29.3, +39.2]	<0.001	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−4.6 [−7.2, −2.0]	0.013	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	+17.0 [+13.6, +20.6]	<0.001	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	−11.1 [−14.1, −8.2]	<0.001	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	+3.0 [+0.7, +5.3]	0.281	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	−1.3 [−2.7, −0.3]	1.000	n.s.	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	−0.3 [−0.9, +0.1]	1.000	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.9, +0.8]	1.000	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+0.3 [−0.1, +0.9]	1.000	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.3, 0.0]	1.000	n.s.	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000	n.s.	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, selective accuracy (conflict)	−1.0 [−4.9, +2.9]	1.000	n.s.	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, selective accuracy (conflict)	+28.2 [+23.0, +33.5]	<0.001	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, selective accuracy (conflict)	+0.3 [−3.3, +3.8]	1.000	n.s.	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, selective accuracy (conflict)	+30.8 [+25.9, +35.6]	<0.001	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, selective accuracy (conflict)	0.0 [−3.9, +3.9]	1.000	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, selective accuracy (conflict)	+15.9 [+11.5, +20.3]	<0.001	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, selective accuracy (conflict)	+0.6 [−2.2, +3.3]	1.000	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, selective accuracy (conflict)	+23.2 [+18.9, +27.7]	<0.001	significant	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, selective accuracy (conflict)	−0.1 [−3.9, +3.8]	1.000	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, selective accuracy (conflict)	+24.2 [+19.7, +29.0]	<0.001	significant	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, selective accuracy (conflict)	+1.6 [−1.6, +4.8]	1.000	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, selective accuracy (conflict)	+26.4 [+21.9, +31.0]	<0.001	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, selective accuracy (conflict)	−0.3 [−4.0, +3.3]	1.000	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain,	+14.7 [+10.3, +19.0]	<0.001	significant	

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, selective accuracy (conflict)	+3.9 [+1.2, +6.7]	0.045	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, selective accuracy (conflict)	+21.1 [+16.9, +25.4]	<0.001	significant	–
H24: detector versus generator reliability (crossed)					
H24	gain rises with detector-minus-generator reliability, Spearman over 30 cells	$\rho = +0.64$	<0.001	supported	–
H24	per-question slope of gain on reliability gap, within cells	+47.6 [+43.2, +51.9]	<0.001	supported	–

Table 12: Pre-registered tests added in the fourth and fifth (H20–H26) (continued), each committed before the runs it covers. Same conventions as Table 4. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

		Plain	Oracle, conflict q.			LLM, directional flag		LLM, remove suspect	
Generator	Skeleton	conflict	flag	dir. flag	remove	conflict	clean	conflict	clean
gpt-4o-mini	default	22.1	**−6.7**	**+13.4**	**+27.6**	+12.5	−3.4	+18.7	**−20.6**
gpt-4o-mini	minimal	20.4	**−5.3**	+5.0	**+18.9**	+3.6	−10.3	+11.0	**−19.6**
gpt-4.1	default	21.8	**−8.6**	**+18.2**	**+30.2**	+8.4	0.0	**+20.6**	**−24.3**
gpt-4.1	minimal	33.6	**−11.0**	**+12.7**	**+14.9**	–	–	–	–
Qwen2.5-7B	default	19.4	−5.8	+3.6	**+18.0**	+2.9	−7.1	+7.9	**−18.3**
Qwen2.5-7B	minimal	17.3	−5.8	+2.9	**+10.8**	–	–	–	–
gpt-6-luna‡	default	14.4	**−3.1**	**+39.1**	**+38.8**	**+29.7**	**−11.6**	–	–
gpt-6.1-terra‡	default	10.8	−0.7	**+39.6**	**+39.6**	**+38.8**	−0.8	–	–

Table 13: The main effects across generators and prompt skeletons (RAMDocs hold-out, 3 seeds, one-way strict accuracy). *Plain*: accuracy (%) on conflict questions with no flags. Other columns: paired difference from plain in points. *Default* skeleton tells the model to give every valid answer; *minimal* drops that policy (Appendix A). **Bold**: significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). The LLM detector is gpt-4o-mini for every generator. Qwen2.5-7B-Instruct runs locally (4-bit), seed 0. ‡: current reasoning models at their default temperature (gpt-6-luna 3 seeds, gpt-6.1-terra seed 0).

	Pair level			Clean-q.	Direc-	Removal, per question		Latency	Cost
Detector	P	R	F1	flagged	tion	misinfo gone	correct lost	(s)	($/1k q.)
CAAG (heuristic)	21.7	7.9	11.6	34.1	–	9.4	31.7	0.23	0.000
NLI cross-encoder	20.1	29.3	23.8	61.9	–	44.6	52.4	1.18	0.000
LLM (gpt-4o-mini)	37.2	20.5	26.4	76.2	88.3	61.2	65.9	0.80	0.181

Table 14: Detectors on the RAMDocs hold-out split (300 questions; percentages). *Clean-q. flagged*: share of conflict-free questions with at least one flag. *Direction*: share of correctly flagged contradictions whose suspect side is the misinformation. *Misinfo gone*: conflict questions whose misinformation documents are all removed when removing the suspect (or, without direction, the second passage). *Correct lost*: clean questions that lose at least one correct document. Latency is per question; CAAG and NLI run locally.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	35.2	37.2	25.9	49.7	911
Prompt only					
Instruction (v1)	+1.3	+0.5	−1.0	+2.1	948
Instruction (v2)	+2.2	+1.4	+3.6	−1.1	951
Oracle detector					
Oracle, flag	−3.6	−4.0	**−7.7**	0.0	980
Oracle, directional flag	+7.0	**+7.9**	**+15.1**	0.0	998
Oracle, remove suspect	**+15.4**	**+17.5**	**+33.3**	0.0	825
Real detectors (exploratory)					
Random, flag	+0.4	+0.4	−0.2	+1.1	949
CAAG, flag	−0.9	−1.4	−1.9	−0.8	942
CAAG, remove	−4.0	−4.5	−1.0	−8.5	809
NLI, flag	+0.2	0.0	−1.4	+1.6	1021
NLI, remove	−8.0	−7.9	−3.1	−13.2	592
LLM, flag	−3.4	−3.4	−7.2	+0.8	970
LLM, directional flag	+4.7	+4.5	+14.1	−6.1	983
LLM, remove suspect	−2.1	−2.6	+18.9	**−26.5**	718
Reference					
MADAM-RAG	+5.7	+5.2	+3.6	+6.9	8196

Table 15: RAMDocs hold-out, gpt-4o-mini, 3 seeds, bidirectional strict accuracy (robustness). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

	Test (labels: *contains*)	One-way [95% CI]	p_{Holm}	Bidir.	p_{Holm}
H2	oracle+flag < plain (conflict)	−4.6 [−8.1, −1.5]	0.006	−5.2	0.003
H3	oracle+flag_directional > plain (conflict)	+11.6 [+5.6, +17.3]	<0.001	+13.3	<0.001
H3	oracle+flag_directional > oracle+flag (conflict)	+16.2 [+10.8, +21.8]	<0.001	+18.5	<0.001
H3	oracle+remove_suspect > plain (conflict)	+25.8 [+19.5, +32.2]	<0.001	+30.3	<0.001
H4	harmful cells, clean q. (dir. flag; removal); ρ	0/9; 8/9; $\rho = -0.08$ [-0.56, -0.02]			

Table 16: Label robustness: pre-registered tests with every suspect document treated as misinformation (173 conflict and 127 clean hold-out questions).

CAAG setting	θ_s	θ_c	P	R	F1	Clean FPR
Default	0.65	0.35	21.7	7.9	11.6	34.1
Best pair F1	0.40	0.10	20.7	49.6	29.2	87.3
Best question F1	0.55	0.15	22.2	28.8	25.1	60.3

Signals (defaults)	Pair F1	Flag rate	Clean FPR
all signals	11.6	6.2	34.1
negation only	0.6	0.4	4.0
entity only	8.6	4.3	23.0
temporal only	0.0	0.0	0.0

Table 17: CAAG on the RAMDocs hold-out (%). Top: thresholds from a sweep over 121 settings ($\theta_s \in [0.40, 0.90]$, $\theta_c \in [0.10, 0.60]$). Bottom: each signal alone.

Condition	All	Conflict	No conflict
No flags (plain)	65.8	46.9	73.6
Instruction (v2)	+0.6	+3.9	−0.8
CAAG, flag	+1.1	+2.4	+0.6
NLI, flag	−0.3	−0.5	−0.2
LLM, flag	+**2.7**	+5.3	+1.6
LLM, directional flag	+0.3	+1.4	−0.2

Table 18: CONFLICTS, 237 automatically scorable questions (69 conflict), gpt-4o-mini, one seed; exploratory. Contains-match accuracy (%) and differences in points; no difference is significant (bold would mark raw $p < 0.05$). CAAG: flags 94.2% of conflict and 92.9% of clean questions; NLI: flags 52.2% of conflict and 43.5% of clean questions; LLM: flags 98.6% of conflict and 83.9% of clean questions.

Condition	All	Lab.	Conflict	Clean
No flags (plain)	32.7	34.5	21.8	48.4
Instruction (v2)	+5.8	+5.2	+12.2	−2.6
Oracle, flag	−4.0	−4.5	−**8.6**	0.0
Oracle, directional flag	+**8.4**	+**9.6**	+**18.2**	0.0
Oracle, remove suspect	+**14.0**	+**15.8**	+**30.2**	0.0
LLM, directional flag	+5.4	+4.4	+8.4	0.0
LLM, remove suspect	+1.0	−0.8	+**20.6**	−**24.3**
NLI, remove	−7.0	−7.3	+0.2	−**15.6**

Table 19: Replication with gpt-4.1 as generator (RAMDocs hold-out, 3 seeds, one-way strict accuracy; detector fixed to gpt-4o-mini). Same conventions as Table 3.

Condition	gpt-4o-mini	gpt-4.1
No flags (plain)	14.0	29.0
Instruction (v1)	+**8.7**	–
Instruction (v2)	−**6.7**	–
Oracle, flag	+**6.5**	+**19.0**
CAAG, flag	+0.7	–
NLI, flag	+3.2	–
LLM, flag	+**6.2**	–

Table 20: WikiContradict (253 genuine conflicts between two Wikipedia passages; the reference answer is both sides), 3 seeds (gpt-4.1: the oracle flag only). First row: share of answers giving both sides (%); other rows: difference in points, **bold** = significant under the paper's single scheme (H8 is a registered family; the gpt-4.1 column is exploratory).

	RAMDocs, gpt-4o-mini			RAMDocs, gpt-4.1		ConflictBank		CB majority		
Method	Lab.	Conflict	Clean	Lab.	Conflict	Conflict	Clean	Conflict	Clean	Calls
No flags (plain), accuracy	31.3	22.1	41.5	32.8†	19.4†	51.1	89.0	21.6	89.0	1
Instruction (v2) only	+2.3	+5.3	−1.1	–	+12.2	–	–	–	–	1
Oracle, flag + instr.	−1.8	−2.4	−1.1	+1.5†	+5.8†	–	–	–	–	1
Oracle, dir. flag + instr.	+6.4	+13.2	−1.1	**+14.7**†	**+30.9**†	–	–	–	–	1
MADAM-RAG	+5.7	+3.6	+7.9	–	–	–	–	–	–	9×tok.
Astute RAG	**−15.6**	−8.4	**−23.5**	–	–	−2.1	−9.7	+1.4	−9.7	2
Astute RAG (multi)	−5.9	−1.2	−11.1	+2.3†	+13.7†	–	–	–	–	2
TrustRAG	**−17.5**	−11.3	**−24.3**	–	–	−9.7	−4.0	+6.8	−4.0	3
TrustRAG (multi)	**−18.1**	−12.2	**−24.6**	−10.9†	0.0†	–	–	–	–	3
Single call (name suspects, then answer)	+6.2	+12.0	−0.3	–	–	+7.6	−1.0	+6.8	−1.0	1
LLM detector, flag + instr.	−2.0	−2.2	−1.9	–	–	–	–	–	–	1+1
LLM detector, dir. flag + instr.	−3.5	+8.4	**−16.7**	–	–	–	–	–	–	1+1
LLM detector, dir. flag	+4.9	+12.5	−3.4	+8.7†	+14.4†	+8.4	−4.7	**+5.7**	−4.7	1+1
LLM detector, remove suspect	0.0	+18.7	**−20.6**	–	**+20.6**	+7.9	−7.0	+7.0	−7.0	1+1

Table 21: Resolution instructions, published baselines and a single-call baseline. First row: accuracy (%, one-way strict) with no flags; other rows: paired difference from it in points. RAMDocs uses the multi-answer prompt (it has same-name questions with several answers); ConflictBank and its majority variant (three wrong-side passages, one correct) use the single-answer prompt and gpt-4o-mini. *Lab.*: labelled RAMDocs questions. 3 seeds, except †: gpt-4.1 seed 0 only, relative to the seed-0 no-flag accuracy in the first row (other gpt-4.1 cells are relative to its 3-seed no-flag accuracy, Table 19). *instr.*: the v2 reliability instruction. *(multi)*: the baseline's final step asks for every valid answer. *Single call*: one call that names suspect passages, then answers. Calls per question; *1+1*: one detector call and one generator call. **Bold**: significant under the paper's single scheme.

Detector	Generator	Dir. acc.	Gen. acc.	Wrong	Gap	Score	Gain
gpt-6.1-terra	gpt-6.1-terra	29.7	73.0	4.3	−43.3	+2.7	+2.7
gpt-6.1-terra	Claude Haiku 5.5	29.7	66.8	4.3	−37.1	+4.3	**+4.1**
gpt-6.1-terra	gpt-4.1	29.7	64.7	4.3	−35.0	+4.7	**+5.0**
gpt-6-luna	gpt-6.1-terra	45.7	73.0	6.7	−27.3	+5.3	+2.7
gpt-6.1-terra	gpt-6-luna	29.7	53.6	4.3	−23.9	+7.7	**+7.3**
gpt-6-luna	Claude Haiku 5.5	45.7	66.8	6.7	−21.1	+7.1	**+5.0**
gpt-6.1-terra	gpt-4o-mini	31.3	51.1	4.0	−19.8	+12.8	**+9.6**
gpt-6.1-terra	Qwen2.5-7B	31.0	50.3	4.0	−19.3	+11.7	**+8.0**
gpt-6-luna	gpt-4.1	45.7	64.7	6.7	−19.0	+7.7	**+6.3**
gpt-6-luna	gpt-6-luna	45.7	53.6	6.7	−7.9	+13.8	**+11.1**
gpt-4o-mini	gpt-6.1-terra	67.0	73.0	30.3	−6.0	−4.7	**−11.7**
gpt-6-luna	gpt-4o-mini	45.7	51.1	6.7	−5.4	+19.1	**+13.7**
Claude Haiku 5.5	Claude Haiku 5.5	61.7	66.8	11.3	−5.1	+9.8	**+5.7**
gpt-6-luna	Qwen2.5-7B	48.0	50.3	6.7	−2.3	+19.7	**+12.0**
Claude Sonnet 5.5	Claude Haiku 5.5	65.0	66.8	9.7	−1.8	+11.7	**+8.2**
gpt-4o-mini	Claude Haiku 5.5	67.0	66.8	30.3	+0.2	+1.9	−4.1
gpt-4o-mini	gpt-4.1	67.0	64.7	30.3	+2.3	+4.0	−2.3
gpt-4.1	gpt-6.1-terra	79.0	73.0	8.0	+6.0	+15.7	**+7.3**
Claude Haiku 5.5	gpt-6-luna	61.7	53.6	11.3	+8.1	+19.1	**+14.3**
Claude Haiku 5.5	gpt-4o-mini	61.7	51.1	11.3	+10.6	+21.3	**+13.0**
Claude Sonnet 5.5	gpt-6-luna	65.0	53.6	9.7	+11.4	+22.8	**+18.1**
gpt-4.1	Claude Haiku 5.5	79.0	66.8	8.0	+12.2	+20.2	**+11.0**
gpt-4o-mini	gpt-6-luna	67.0	53.6	30.3	+13.4	+14.7	**+8.6**
Claude Sonnet 5.5	gpt-4o-mini	65.0	51.1	9.7	+13.9	+28.1	**+15.9**
gpt-4.1	gpt-4.1	79.0	64.7	8.0	+14.3	+20.7	**+11.7**
gpt-4o-mini	gpt-4o-mini	67.0	51.1	30.3	+15.9	+17.6	**+8.4**
gpt-4o-mini	Qwen2.5-7B	67.0	50.3	30.3	+16.7	+17.3	**+7.3**
gpt-4.1	gpt-6-luna	79.0	53.6	8.0	+25.4	+30.1	**+21.3**
gpt-4.1	gpt-4o-mini	79.0	51.1	8.0	+27.9	+33.8	**+18.6**
gpt-4.1	Qwen2.5-7B	79.3	50.3	7.7	+29.0	+36.7	**+16.0**

Table 22: Every detector–generator cell of H24 (ConflictBank, one-answer prompt, 300 conflict questions; %). *Dir. acc.*: share of questions on which the detector flags the true contradiction and names its misinformation passage; *Gen. acc.*: the generator's accuracy with no flags; *Wrong*: share of questions on which it marks a correct passage as suspect; *Score* (exploratory): right direction where the generator errs minus a correct passage marked where it is right; *Gain*: the LLM directional flag's change in points. Figure 1 plots the same cells.